Team-Based Learning: Histopathology of the Bladder, Prostate, and Testis

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Learning Objectives

1. Describe and identify the risk factors, gross features, and microscopic histology of urothelial carcinoma (transitional cell carcinoma), including contrasting the common growth patterns of this tumor.
2. Explain the pathogenesis of benign prostatic hyperplasia (BPH); describe and identify the clinical presentation, gross features, and microscopic histology of BPH.
3. Describe and identify the gross features and microscopic histology of prostate adenocarcinoma; discuss the use of the Gleason and grade group systems with respect to grading of prostate cancer.
4. Describe and identify the demographics, gross features, and microscopic histology of testicular germ cell tumors (testicular cancer), including the use of serum markers in its diagnosis.
5. Describe the major factors that affect the staging of urothelial carcinoma, prostate cancer, and testicular cancer.

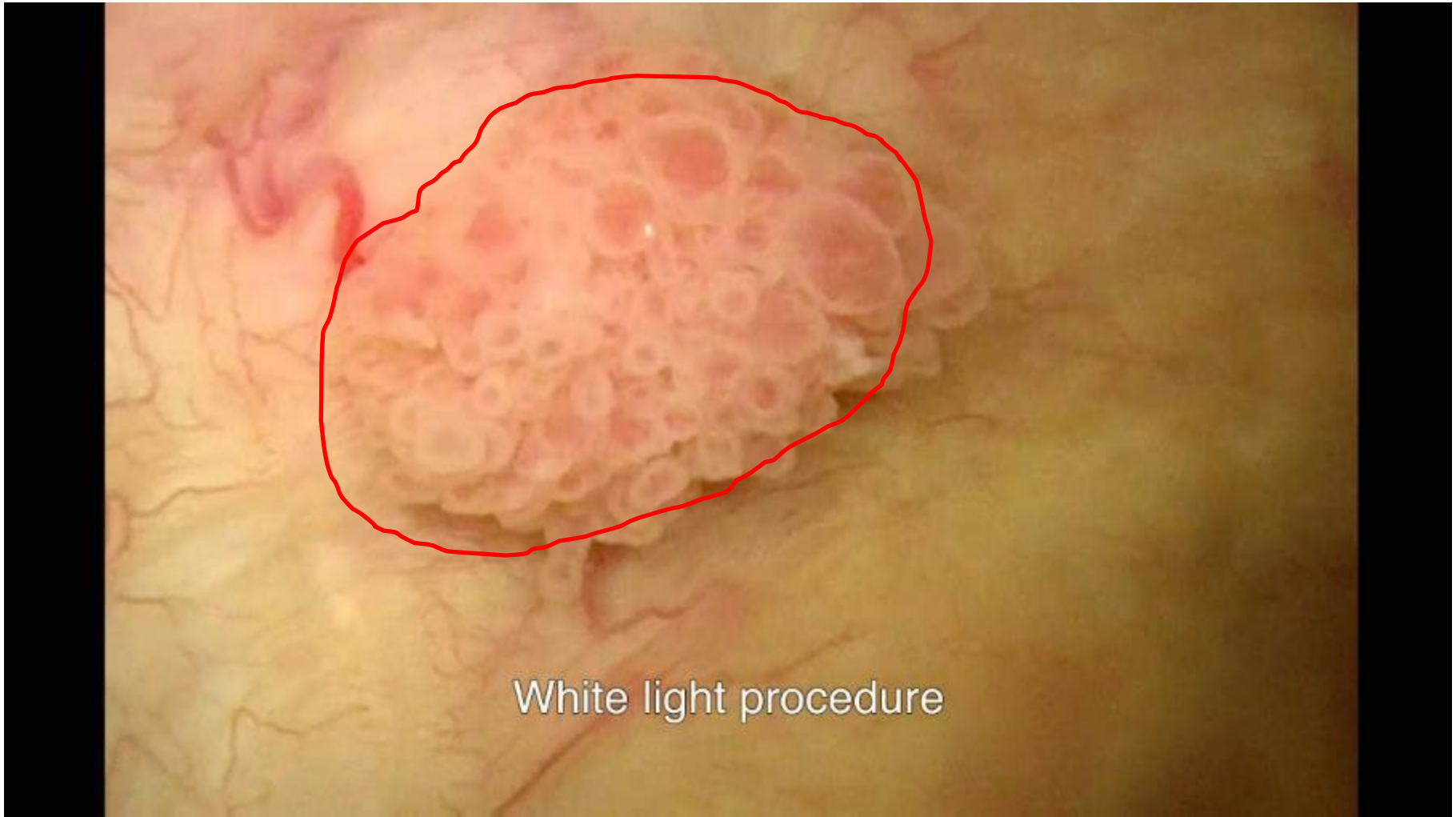
CASE 1

CASE 1 - Clinical history

- A 70-year-old Philadelphia man presented with blood in the urine
- There was no dysuria, pain or increased frequency of urination
- History was remarkable for 50 pack-years of smoking
- Office urinalysis confirmed red blood cells in the urine and no inflammation
- The patient was referred to a urologist who advised cystoscopy. The result of the cystoscopy is shown in the image below:

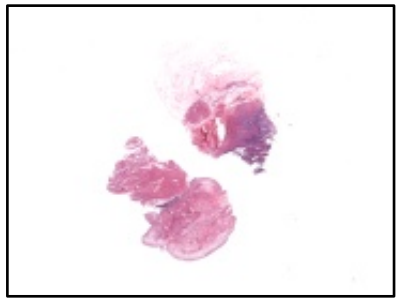
CASE 1 - Cystoscopy of bladder mucosa

A single exophytic tumor was observed on the left lateral wall. Note the papillary fronds

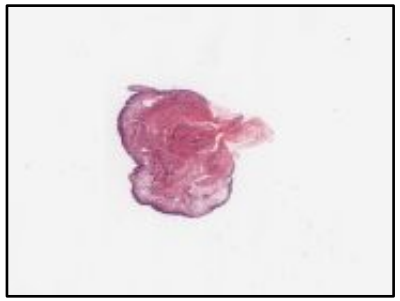


Case 1

The exophytic tumor is resected and sent for evaluation. As the genitourinary pathologist on service, you review the slide linked below:



Bladder, Prostate, Testis VS 1-1



normal for comparison
(no annotations)

Case 1

Examine Bladder, Prostate, Testis VS 1-1 and answer the following questions:

Q1: Describe the appearance of the neoplastic cells, shown in the black circle. From what cell type did they derive?

They are cohesive, which is a feature of epithelial cells.

They derive from the lining cells of the bladder: the urothelium.

Q2: What term would you use for the architectural feature shown in the red rectangle? (Observe the vessel running down the core of this structure, which is covered with neoplastic cells.) Are there any other neoplasms (in other organ systems) you have seen this growth pattern in?

This is a papillary structure. The center is a fibrovascular core, which is the scaffolding for the tumor cells that are proliferating.

Papillary tumors are also seen in the kidney, thyroid, ovary, lung, and other organs.

Case 1

Continue to examine Bladder, Prostate, Testis VS 1-1 and answer the following questions:

Q3: The dashed line approximates the tumor's depth of invasion into the wall of the bladder. Into what layer does the tumor extend? Why is determining the depth of invasion important?

The tumor extends into the lamina propria.

Bladder cancer is staged by depth of invasion.

Q4: What (normal) tissue type is shown within the green circle? What name is given to this tissue within the urinary bladder? The tumor does NOT invade into this tissue type within the specimen; why is that significant?

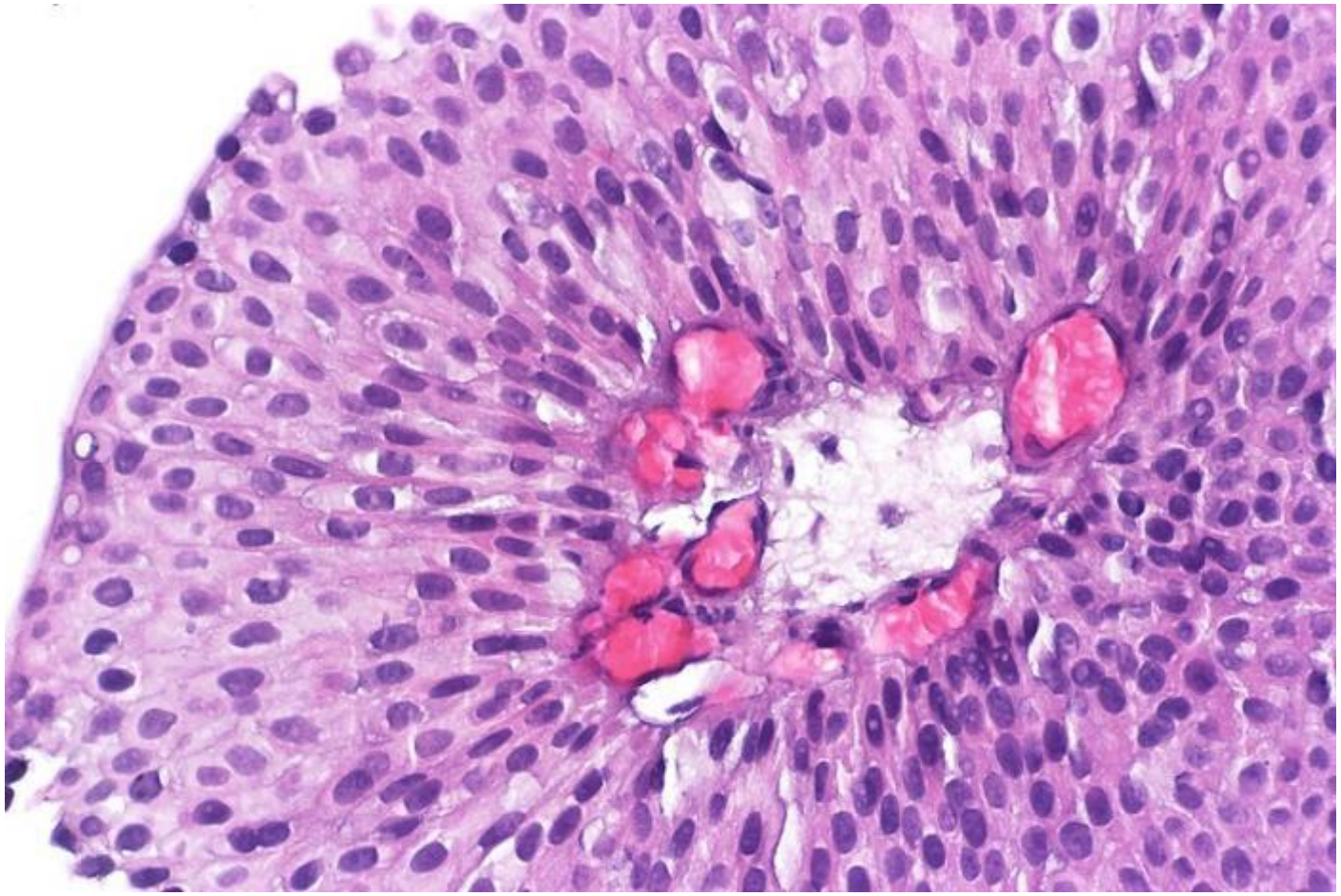
Smooth muscle of muscularis propria/ detrusor muscle

Muscle-invasive bladder cancer has a poorer prognosis and requires more aggressive intervention, including cystectomy.

CASE 1 - Resection of bladder tumor

Low-grade papillary urothelial carcinoma





Urothelial carcinoma is the most common tumor of the bladder

- Urothelial carcinoma can be in situ or invasive, flat or papillary
- They are further classified as either low-grade or high-grade
- Urothelial carcinomas can show areas of squamous or glandular differentiation
- Other epithelial malignancies are rare
  - Primary (pure) squamous carcinomas of bladder and primary (pure) adenocarcinomas of the bladder
  - Neuroendocrine carcinomas
  - Carcinosarcomas

Urothelial neoplasms

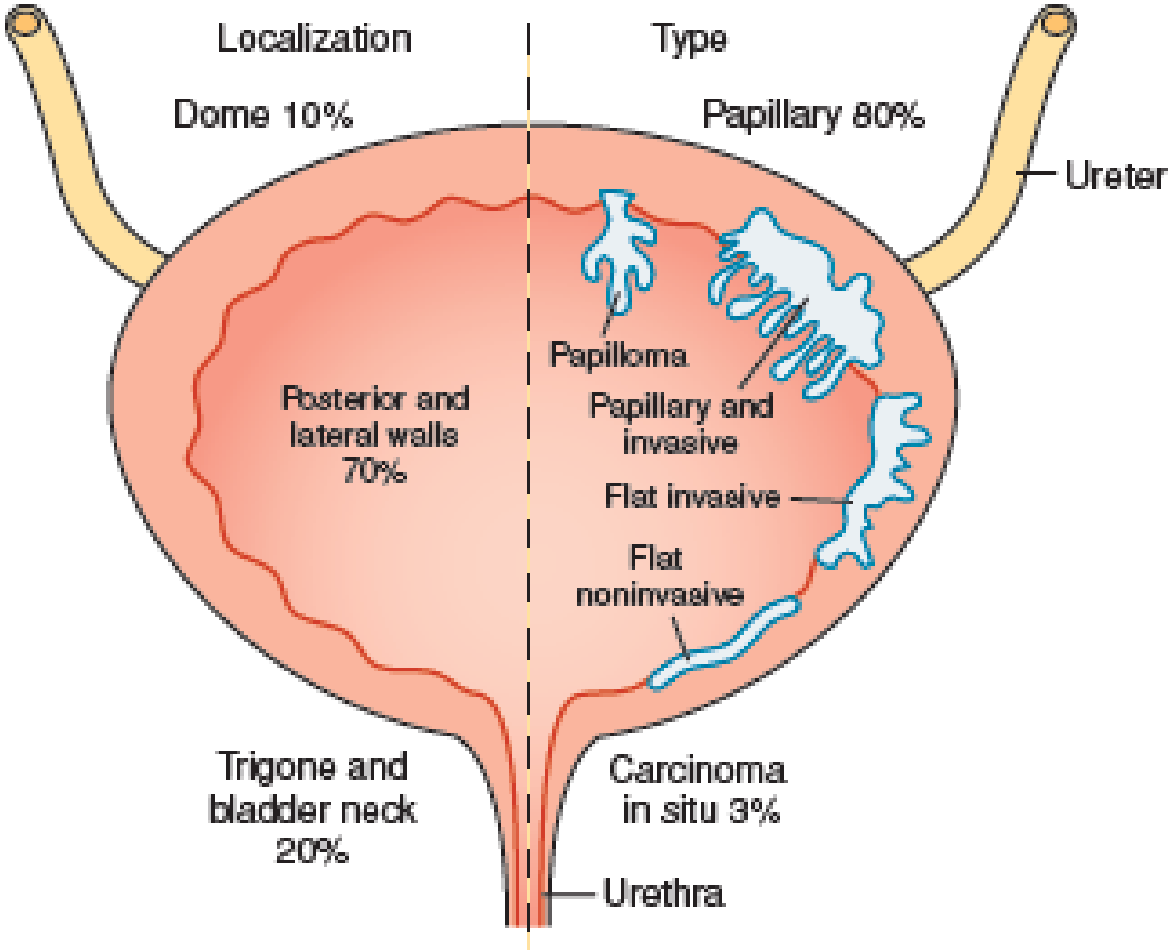


FIGURE 23-16. Urothelial neoplasms. Most tumors arise in the posterior and lateral walls; trigone and bladder neck are involved less often and the dome least. Malignant tumors may be papillary or flat. Both flat and papillary tumors may be invasive or noninvasive. Benign transitional cell papillomas are rare.

UROTHELIAL TUMORS

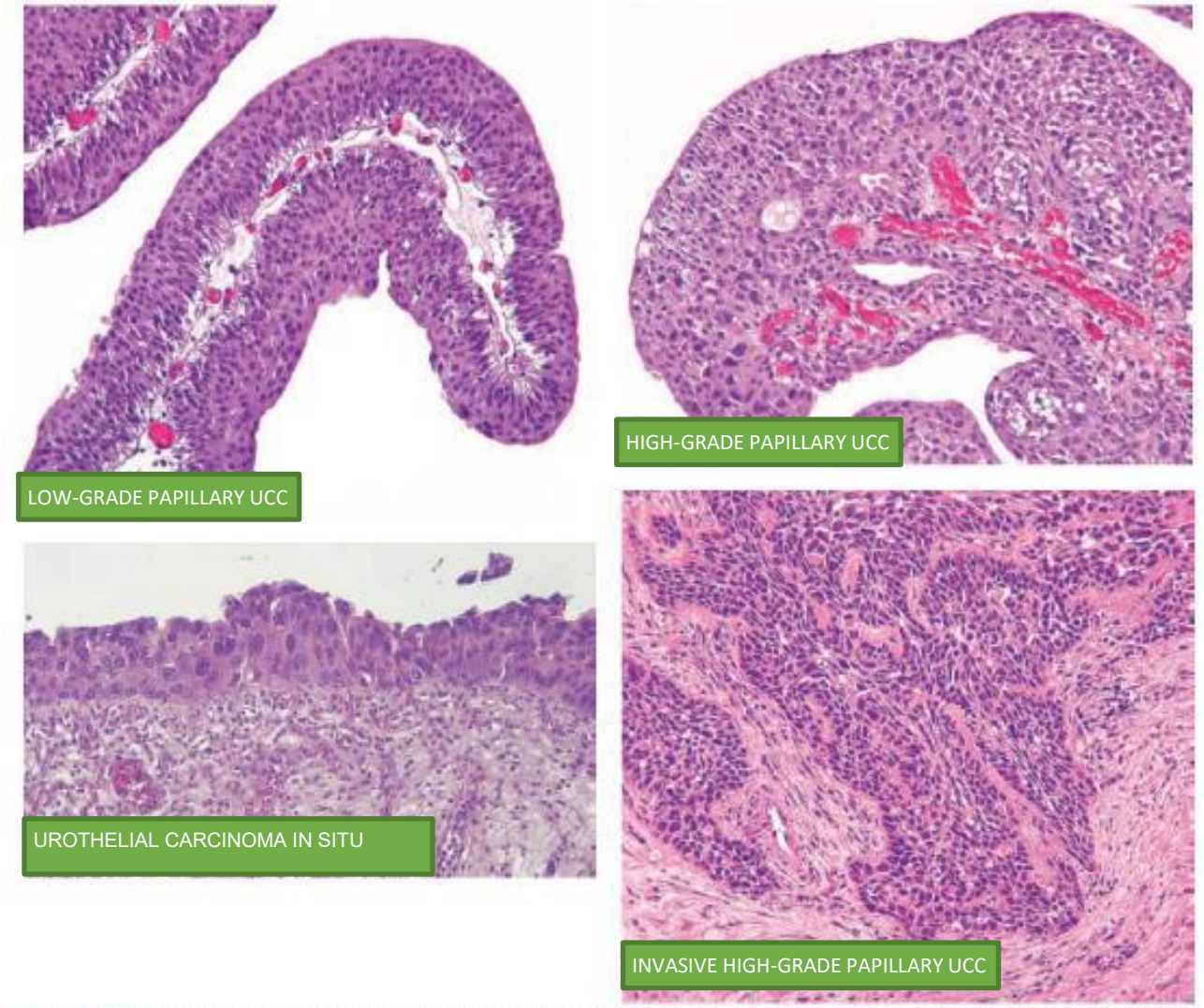
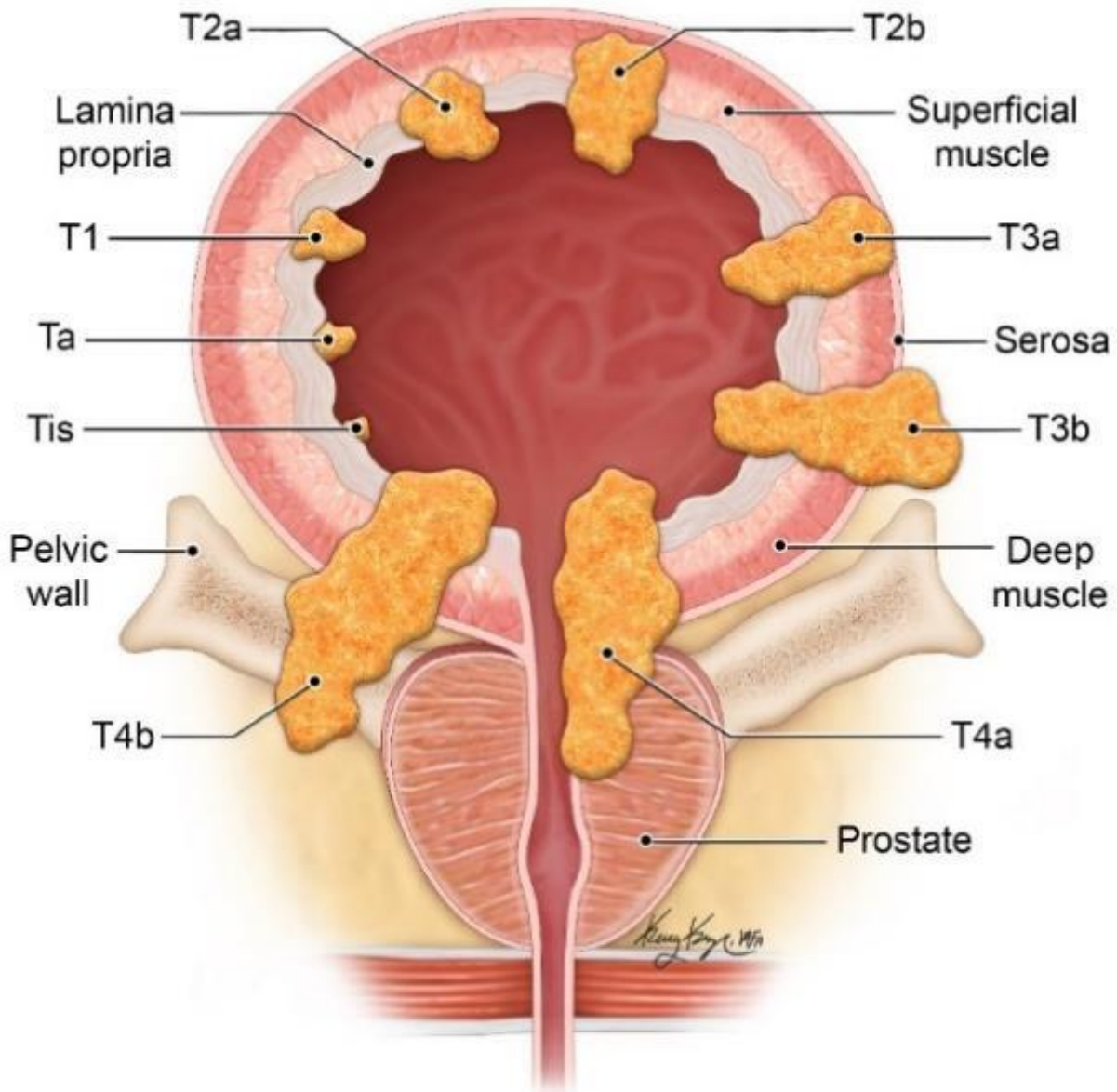


FIGURE 23-18. Urothelial tumors of the urinary bladder. A. Low-grade papillary urothelial carcinoma consists of exophytic papillae that have a central connective tissue core and are lined by slightly disorganized transitional epithelium. **B.** High-grade papillary urothelial carcinoma at higher magnification shows architectural disorder and pleomorphic hyperchromatic nuclei. **C.** Urothelial carcinoma in situ is a flat lesion with marked nuclear pleomorphism and loss of polarity from the basal layer to the surface. **D.** Invasive high-grade papillary urothelial carcinoma consists of irregular nests of hyperchromatic cells invading into the muscularis.

AJCC Cancer Staging



Risk factors for developing bladder tumors

- Smoking, especially cigarettes
- Industrial exposure to aryl amines and related compounds
- Other occupational exposures (e.g., perchloroethylene in dry cleaning)
- Schistosomiasis in endemic parts of Africa where the majority of the cancers are squamous cell carcinomas
- Medications such as long term analgesics or cyclophosphamide therapy
- Irradiation for other pelvic malignancies

Non-epithelial tumors of the bladder are rare

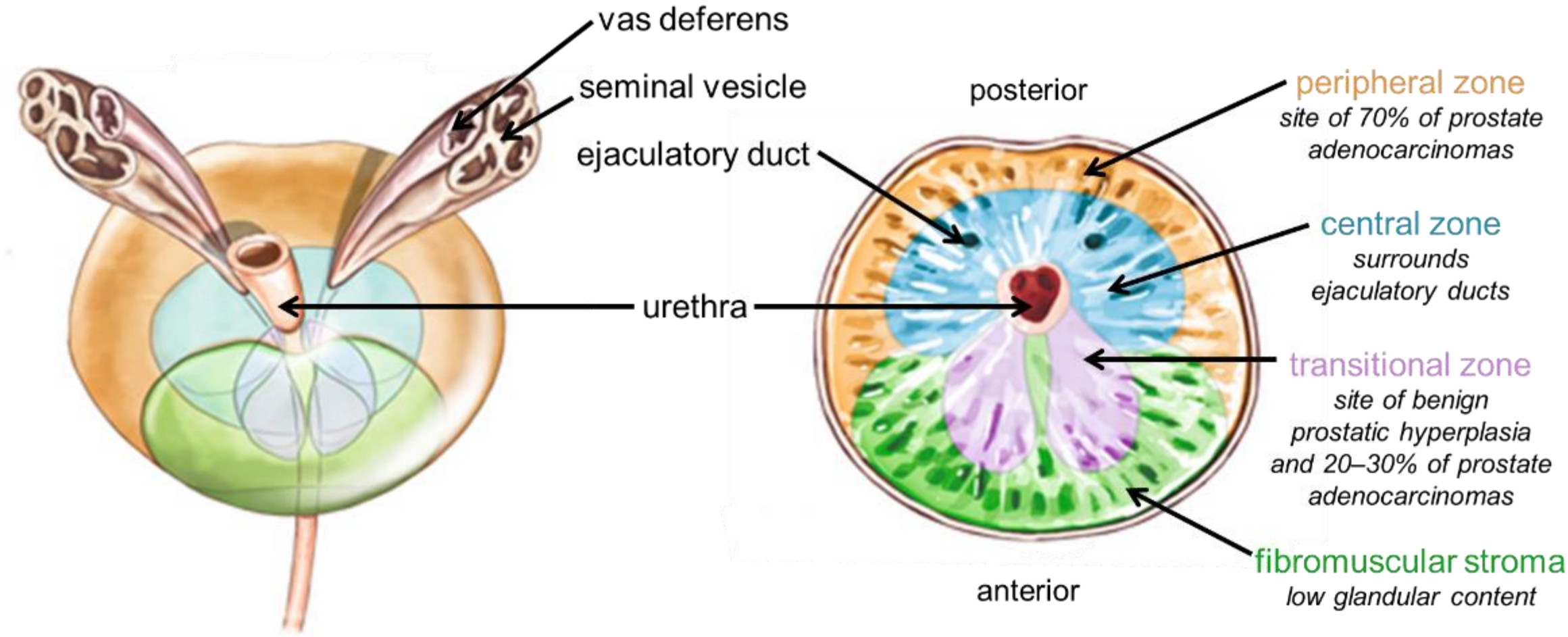
- Non-epithelial tumors include
  - **Benign mesenchymal tumors** such as leiomyoma, solitary fibrous tumor, inflammatory myofibroblastic tumor
  - **Malignant mesenchymal tumors** such as leiomyosarcoma, rhabdomyosarcoma, angiosarcomas
  - Embryonal rhabdomyosarcoma is a tumor in infancy or childhood
  - Lymphomas may involve the bladder in systemic disease
  - Secondary tumors may involve the bladder as direct extension from the cervix, uterus, prostate or rectum

CASE 2

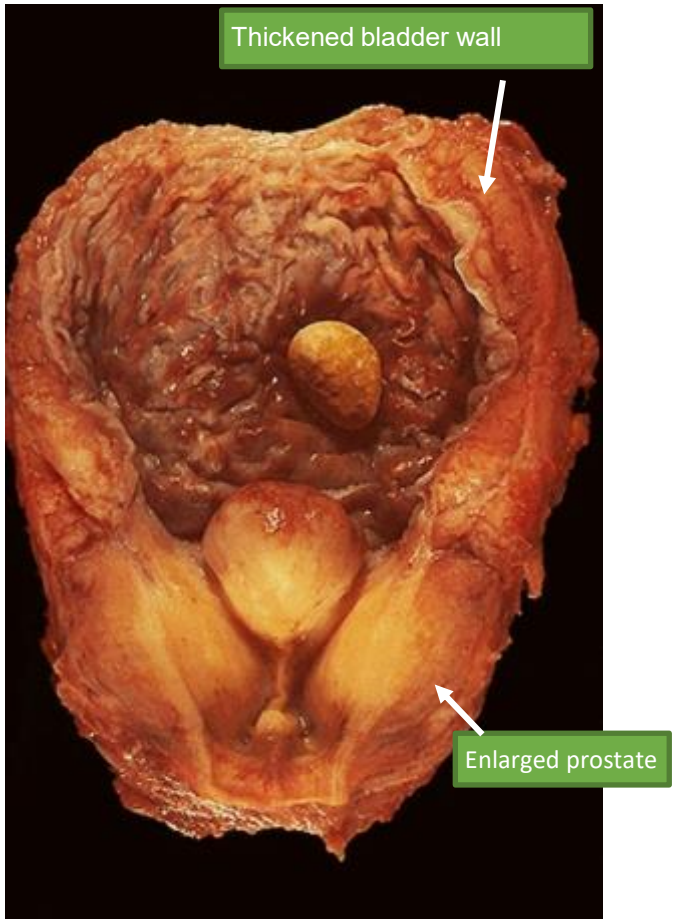
CASE 2 - Clinical history

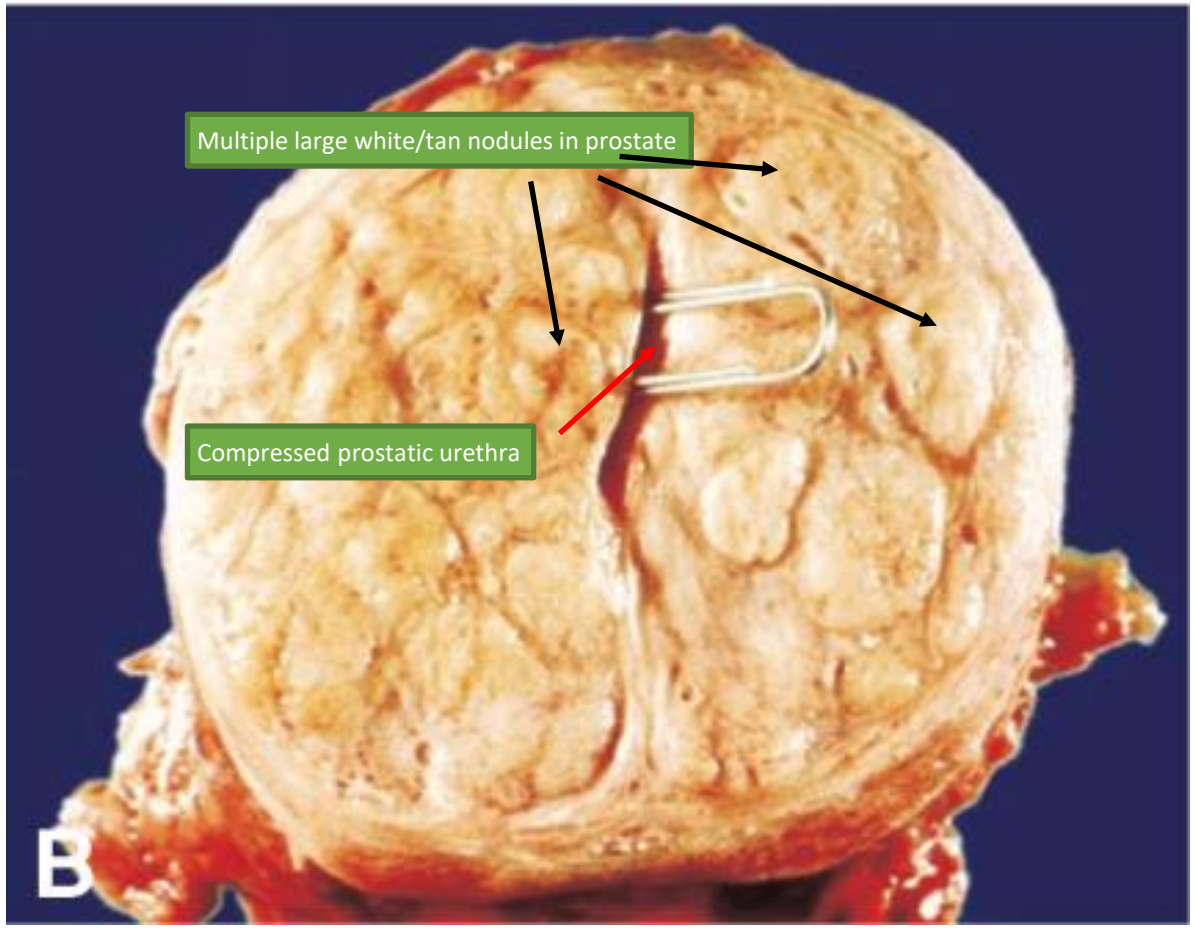
- A 75-year-old man has difficulty with urination
- The patient reports a weak stream, nocturia x 3, and a feeling of never emptying the bladder
- The PSA level is 4 (normal), unchanged for years
- The patient was admitted with a distended bladder, fever, and dysuria; blood culture was positive for *E. coli*, and despite treating the UTI with IV antibiotics, the patient developed septic shock and died
- An autopsy was requested
- Gross images of the bladder and prostate are shown:

Anatomy of the Prostate Gland



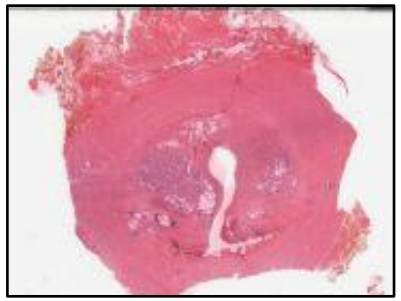
Can you identify the gross pathologic changes from the bladder and prostate?



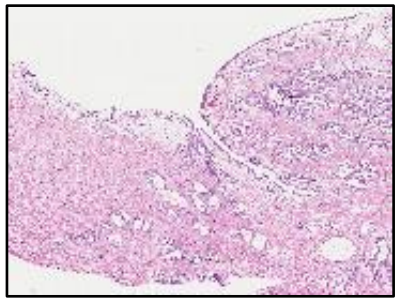


Case 2

This disease is usually a clinical diagnosis, not a pathologic one. However, it may be seen in the background of prostate glands taken out for other reasons (i.e., malignancy). Review the slide linked below:



Bladder, Prostate, Testis VS 2-1



normal for comparison
(no annotations)

Case 2

Examine Bladder, Prostate, Testis VS 2-1 and answer the following questions:

Q1: What structure is represented by the space containing the asterisk? What type of epithelium lines this structure, as highlighted by the black arrows)?

This is the prostatic urethra, lined by urothelium.

Q2: This patient's disease primarily affects the region enclosed with the yellow dashed lines. What is this region of the prostate gland called?

The transitional zone (and perhaps also the central zone)

Case 2

Continue to examine Bladder, Prostate, Testis VS 2-1 and answer the following questions:

Q3: Consider the appearance of the prostatic glands in this patient's specimen on intermediate power, as shown in the blue circle. They are more cellular than normal; what term is used to describe a regulated increase of a certain cell type? Still, they retain the normal architecture of the prostate gland. How might you describe the architecture of these glands? Why is it significant to determine if the normal architecture is maintained?

The regulated increase of a cell type is called hyperplasia.

The architecture of the glands is lobular.

In cancer, the normal lobular architecture is disrupted as the epithelial cells of the tumor break away and become infiltrative.

Q4: The glands have two cell layers, as shown with the white arrows (inner layer) and arrowheads (outer layer). What are the cells in the outer layer called? Why is it important for a pathologist to look for this cell population while evaluating the specimen?

These are basal cells; they are absent in prostate carcinoma.

Case 2

Continue to examine Bladder, Prostate, Testis VS 2-1 and answer the following questions:

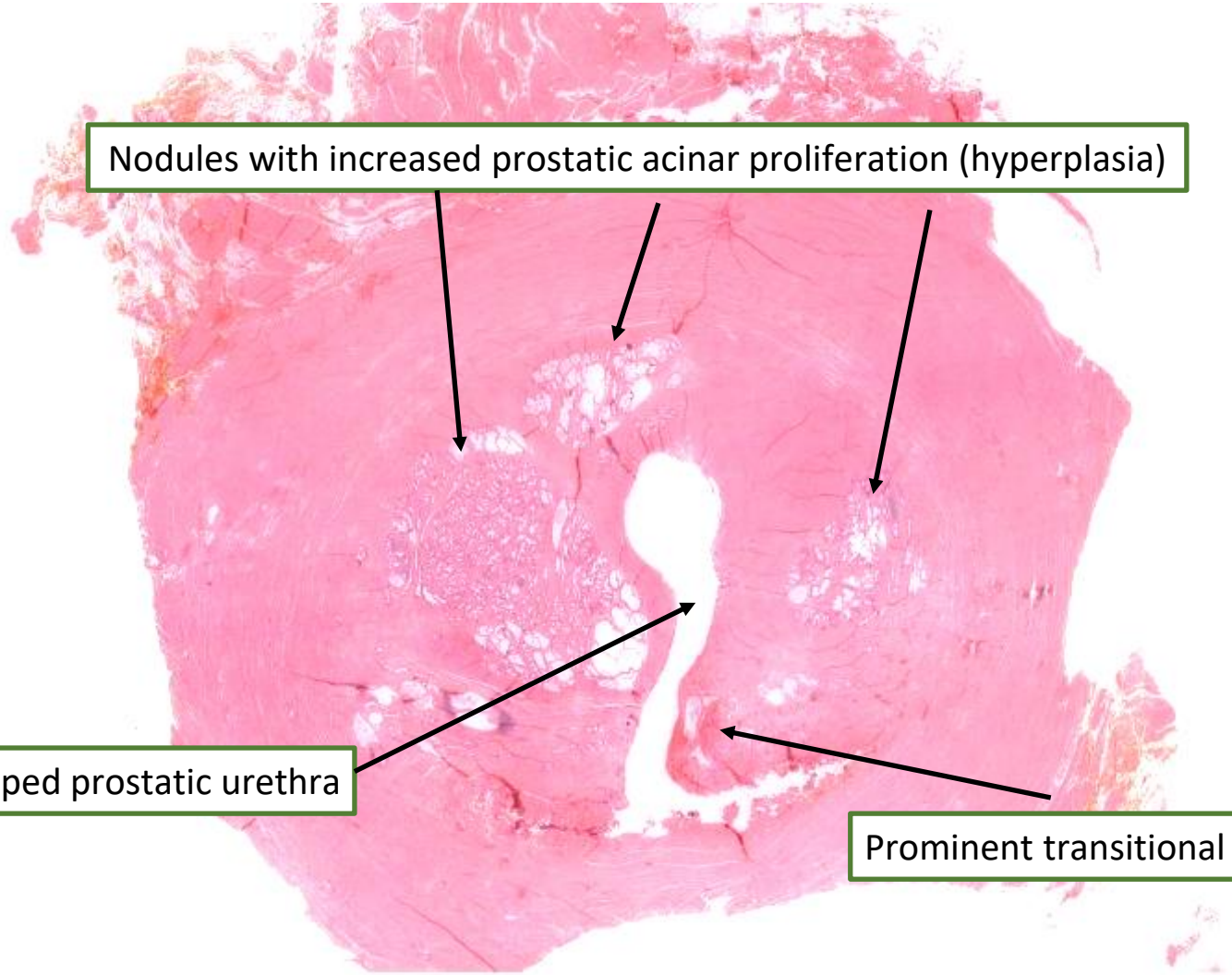
Q5: What are these glassy, layered-appearing structures in the red circles, that are a normal feature of the prostate gland?

These are corpora amylacea; they are seen secondary to cellular desquamation and degeneration.

Q6: How does the connective tissue shown in the green rectangle compare to the normal stroma of the prostate gland? What is this change called?

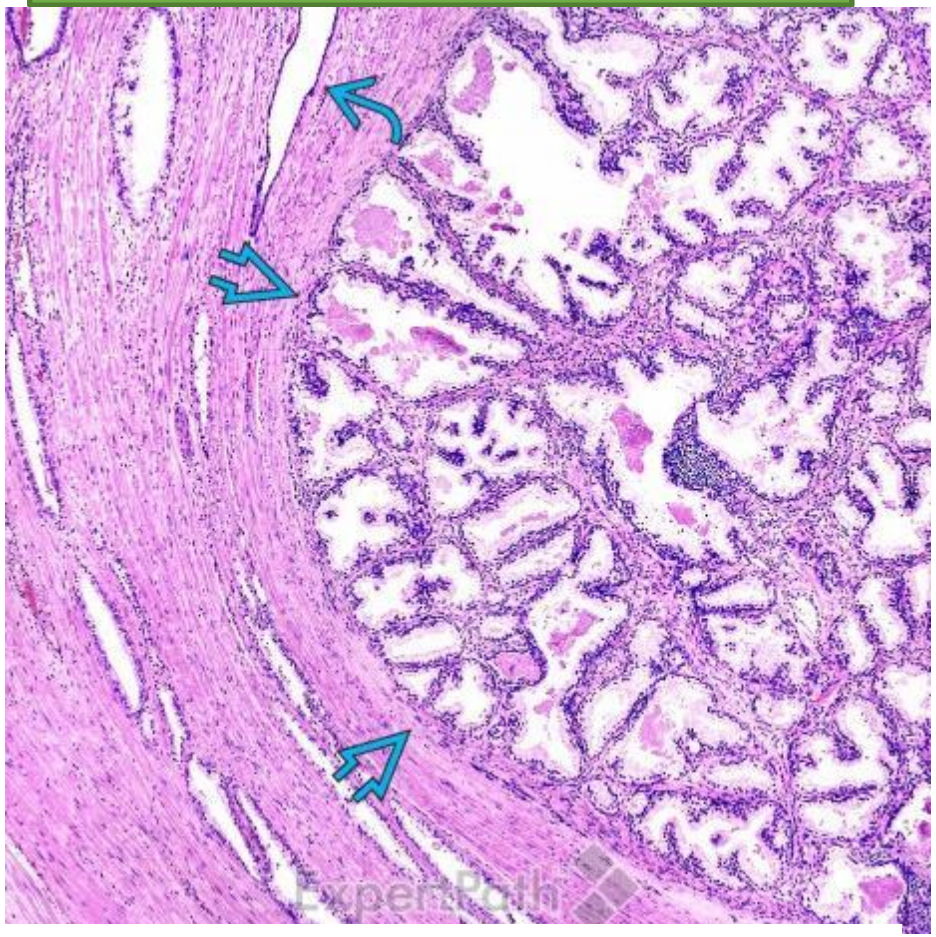
It is more cellular; this is also called hyperplasia.

CASE 2 – Histology of nodular prostatic hyperplasia



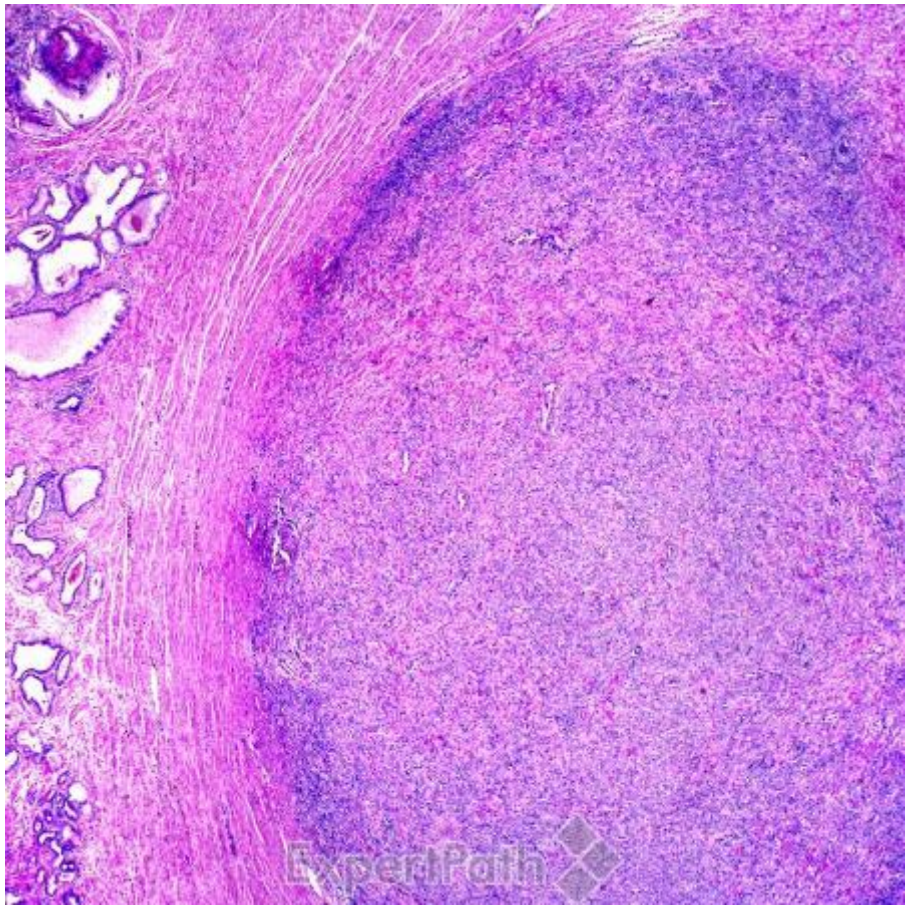
CASE 2: HISTOLOGY OF NODULAR PROSTATIC HYPERPLASIA

Nodule containing hyperplastic prostatic glands



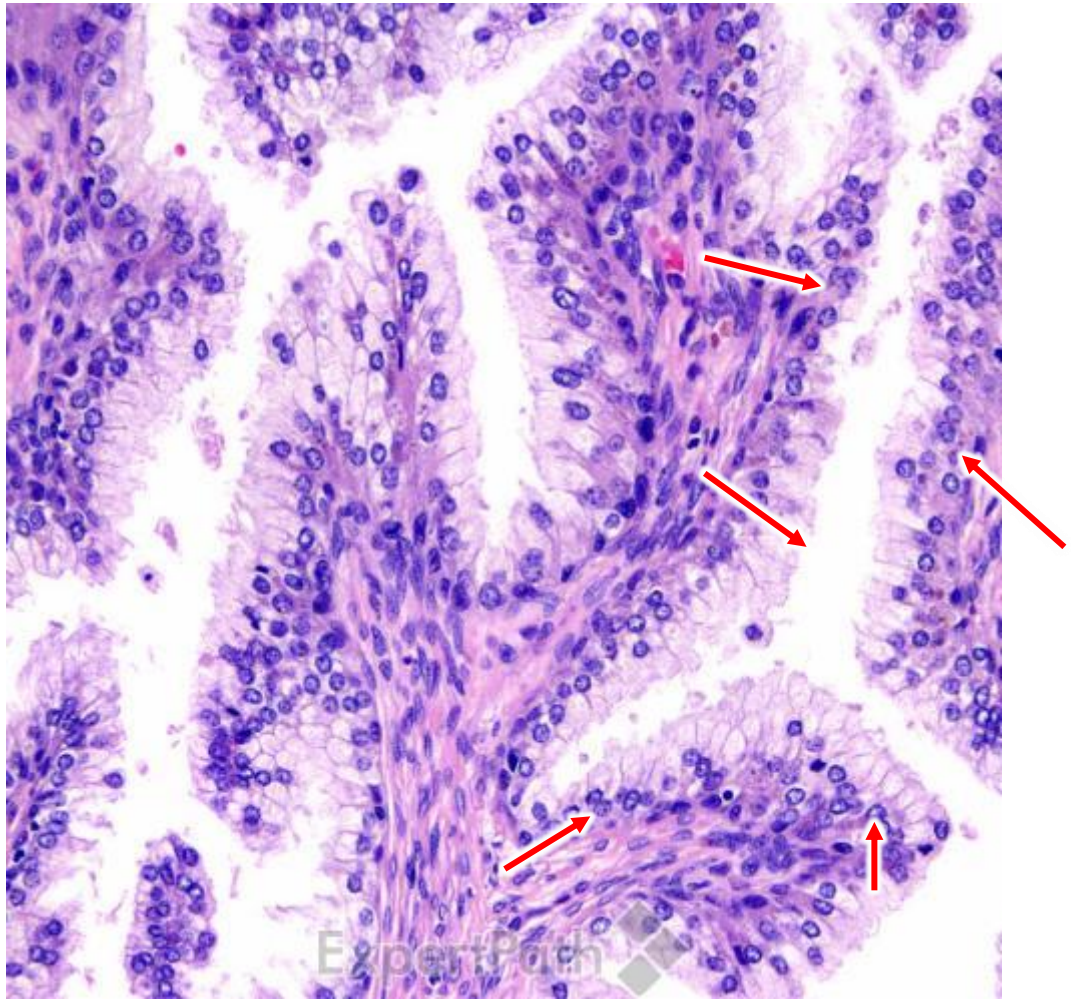
BPH nodule arising from transition zone (cyan open arrow) shows compression with secondary atrophy of the surrounding peripheral zone (cyan curved arrow). The hyperplastic glands are much larger, showing luminal undulations and papillary infoldings, and the hyperplastic stroma is denser compared with peripheral zone.

Nodule of stromal hyperplasia adjacent to hyperplastic glands



Well-circumscribed stromal hyperplastic nodule shows sheets of spindle cells containing abundant capillaries dispersed throughout the stroma.

CASE 2: NODULAR PROSTATIC HYPERPLASIA



Benign hyperplastic glands lined by basal cells
Serrated luminal border

Benign prostatic hyperplasia (BPH) is common

- Leading cause of morbidity in older men
- Seen in 50% of in their 50s, up to 80-90% in their 70s and 80s
- Most common noncancerous cause of serum PSA elevation
- TURP (transurethral resection of the prostate) remains standard in management
- Choice of therapy based on symptoms, prostate size, and existing comorbidities

BPH is hormonal dependent- NOT premalignant

- **Testosterone** is necessary for prostatic development and to maintain secretory function
- The active androgen form, **dihydrotestosterone (DHT)**, is formed from testosterone by the enzyme **5α-reductase** (type 2 from stromal cells and type 1 from skin and liver)
  - DHT binds to androgen receptors (ARs) in glandular and stromal cells and promotes growth of prostatic glandular and stromal tissue
  - DHT induced growth factors increase proliferation of stromal cells and decrease epithelial cell death
  - Prostate hyperplasia is rare in patients who have undergone castration before puberty
  - Some, but not all, patients will respond to medical therapy that blocks **dihydrotestosterone (DHT) and 5α-reductase**

Therapeutic modalities for BPH are numerous

- Symptomatic relief may be obtained by alpha-adrenergic blockers through reducing the tone of smooth muscle in the prostate
- 5α-reductase inhibitors shrink the size of the prostate by decreased DHT synthesis
- Transurethral resection of the prostate was the gold standard
- Tissue destructive methods include cryotherapy, heat therapy, and radioablation

CASE 3

CASE 3 – Clinical history

- A 62-year-old man has well-controlled hypertension and no other major medical complaints
- Routine blood work has shown a rising PSA over the course of 3 years
- On digital rectal exam by the urologist, a distinct firmness along the left side of the prostate
- Transrectal biopsies were performed and the diagnosis was adenocarcinoma
- The patient opted for a radical prostatectomy for definitive therapy

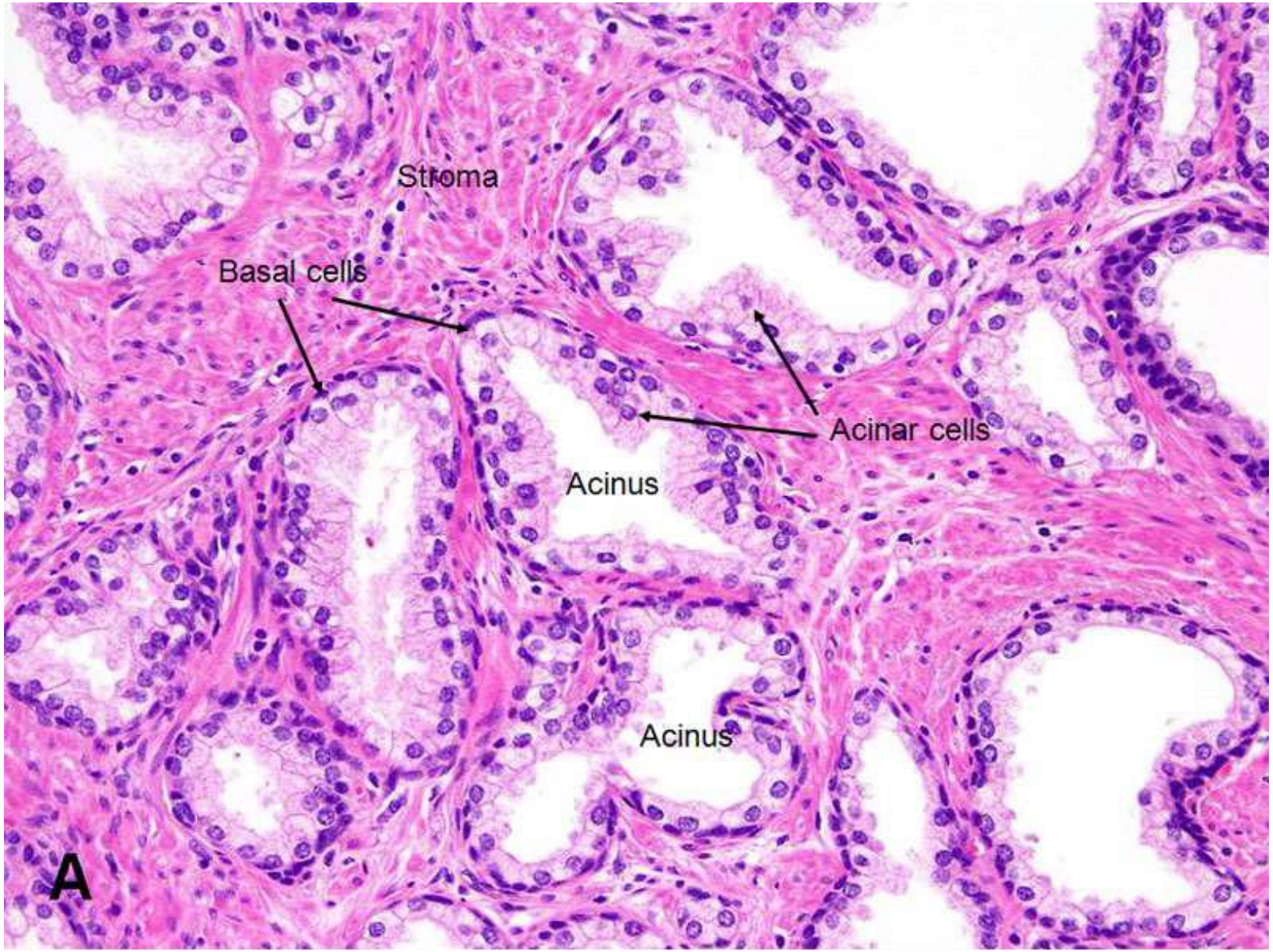
CASE 3 - Prostatectomy



PCa shows a relatively dense, homogeneous solid area (white open arrow) at the posterolateral aspect of the peripheral zone. PCa most commonly occurs at this site. Most PCas are multifocal.

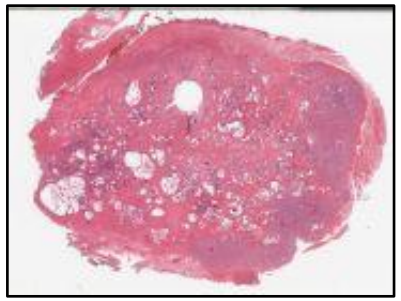
Normal benign prostatic acini

- The prostate is composed of glandular elements (known as acini) embedded in a fibromuscular stroma
- Benign acini are lined by a peripheral layer of basal cells
- Acinar cells are typical tall columnar cell with small, basally located nuclei
- Nucleoli are inapparent

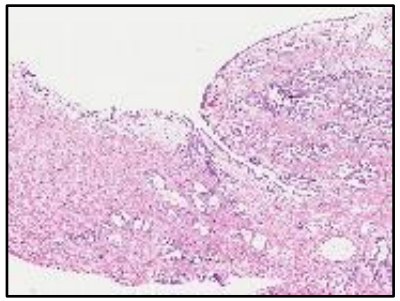


Case 3

The patient undergoes a radical prostatectomy. As the genitourinary pathologist on service, you review the slide linked below:



Bladder, Prostate, Testis VS 3-1



normal for comparison
(no annotations)

Case 3

Examine Bladder, Prostate, Testis VS 3-1 and answer the following questions:

Q1: Look at the neoplastic glands shown in the black box. How many layers of cells line each gland?

Just one layer (normal = two layers).

Q2: Some of the neoplastic glands have viscous, lavender-blue material within their lumen, as shown with the green arrows. What is this material?

"Blue mucin," usually associated with malignant glands.

Q3: The vast majority of this patient's neoplasm is composed of small, well-formed glands with intervening stroma (connective tissue), as shown in the blue box. What system is used to describe the formation of glandular structures in this patient's malignancy? Why do pathologists grade these patterns?

The Gleason grading system; this determines differentiation and correlates with behavior of the tumor.

Case 3

Continue to examine Bladder, Prostate, Testis VS 3-1 and answer the following questions:

Q4: In some areas, like the one indicated with the red box, there are back-to-back glands without intervening stroma that form sieve-like structures. What term is used to describe this “sieve-like” appearance? (It was also used to describe the architecture of some cases of DCIS in the breast.) While you won't be expected to grade the patterns of this type of cancer, do (a) sieve-like, back-to-back glands or (b) well-formed glands with intervening stroma carry a higher number in the Gleason grading system?

> *“Cribriforming” describes this sieve-like pattern; cribriforming glands carry a higher grade (grade 4) than well-formed ones with intervening stroma (grade 3).*

Q5: What is the structure indicated by the yellow arrow? Given the relationship of the neoplastic gland to this structure, what phenomenon is shown within the yellow box? What is the significance of this finding?

> *A nerve with tumor wrapping around it, which can be described as perineural invasion. Perineural invasion is correlated with extraprostatic extension of the tumor.*

Case 3

Continue to examine Bladder, Prostate, Testis VS 3-1 and answer the following questions:

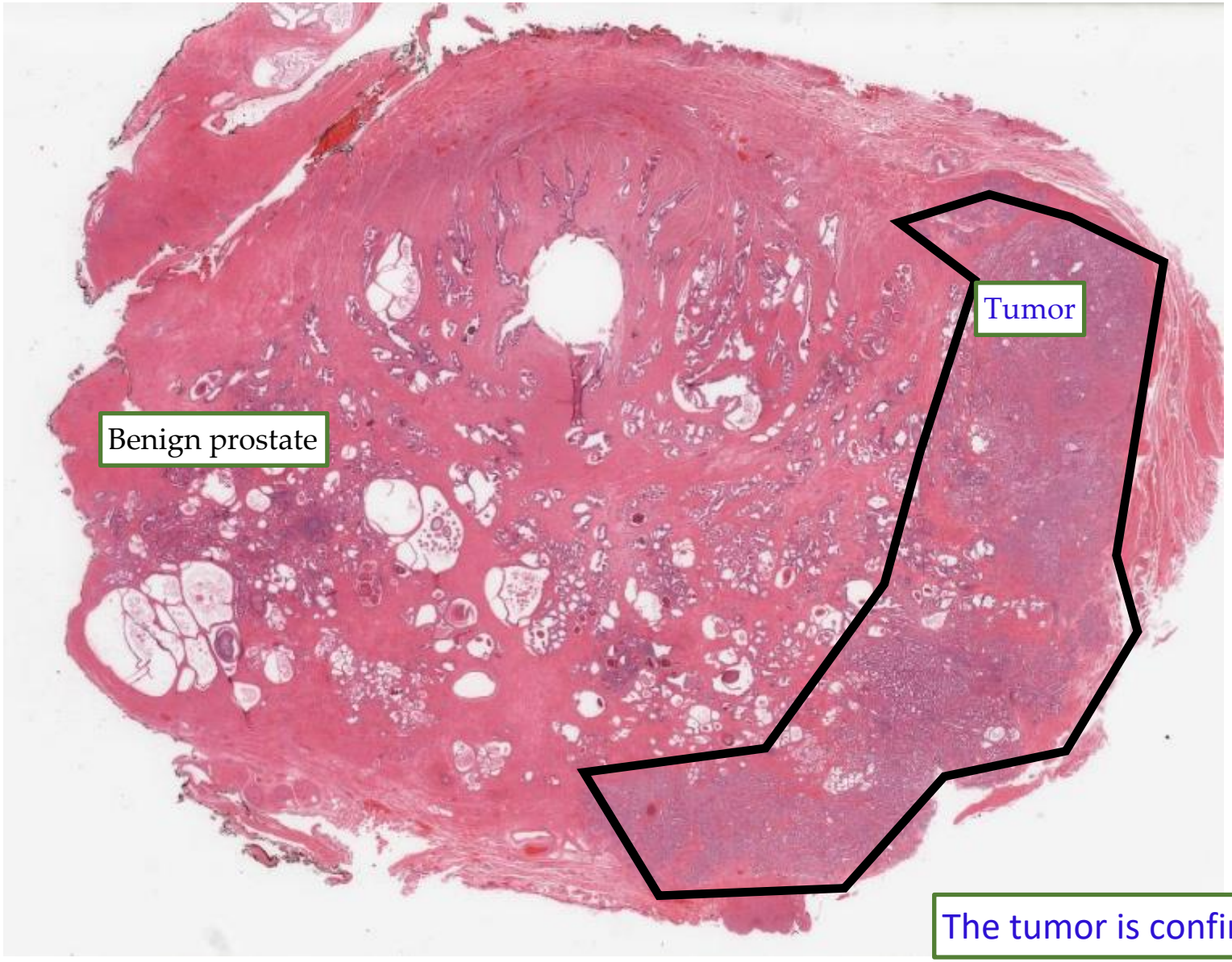
Q6: The patient's neoplasm does not breach the dense regular connective tissue indicated by the white arrows. What is this fibrocollagenous layer that surrounds the prostate gland? Why is it significant that the patient's neoplasm does NOT cross this tissue plane?

> *This is the prostatic capsule. If the tumor breached this layer, it would constitute extraprostatic extension and would be a higher stage in the TNM staging system.*

Q7: What is the black material indicated by the black arrows? How does this material assist in the pathologist's evaluation of the specimen?

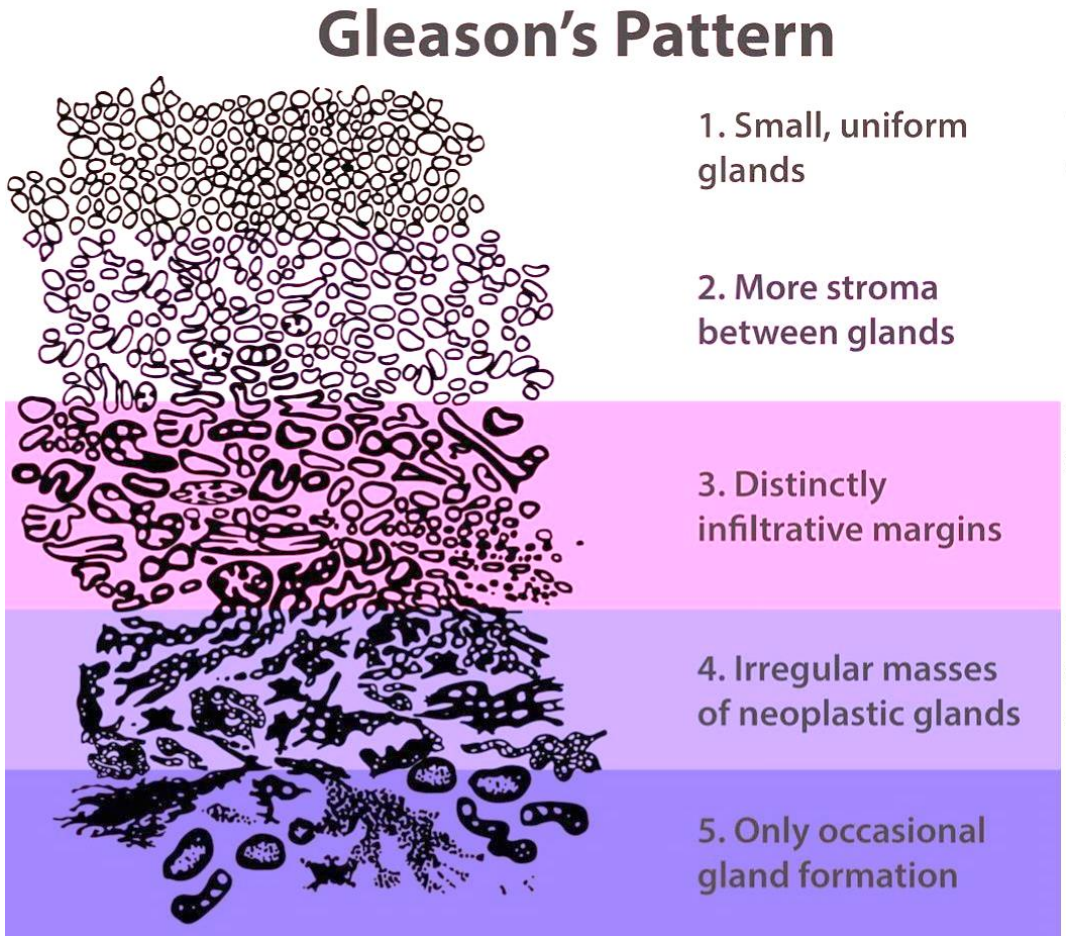
> *It is ink, indicating the margin of resection. The ink identifies the surgical border of the specimen, which must be evaluated to see if the tumor extends to the margin. This prostatectomy has negative margins of resection.*

CASE 3 - Whole mount section of prostatectomy



The tumor is confined within the prostatic capsule

CASE 3 - grading of prostatic adenocarcinoma
Gleason grading



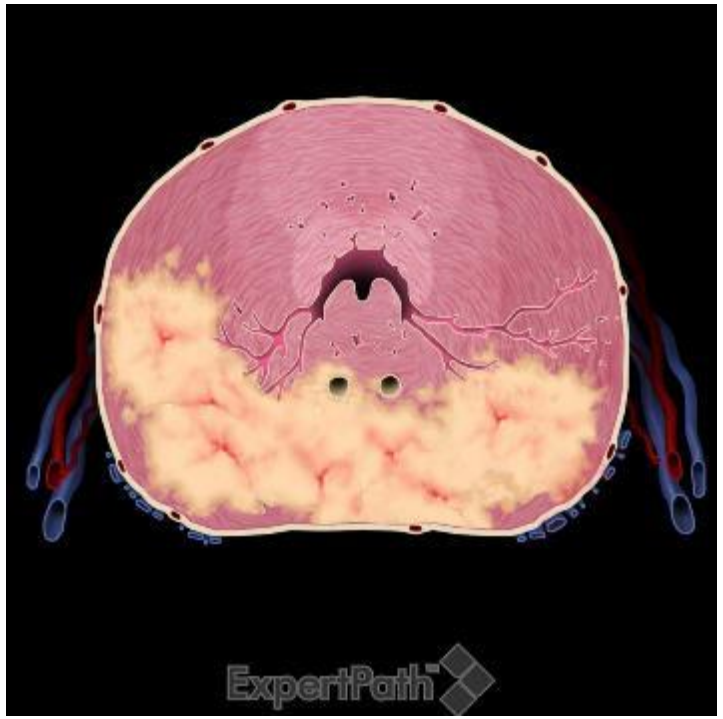
Patterns 1, 2, 3 - separated, well formed glands

Pattern 4 - complex glands

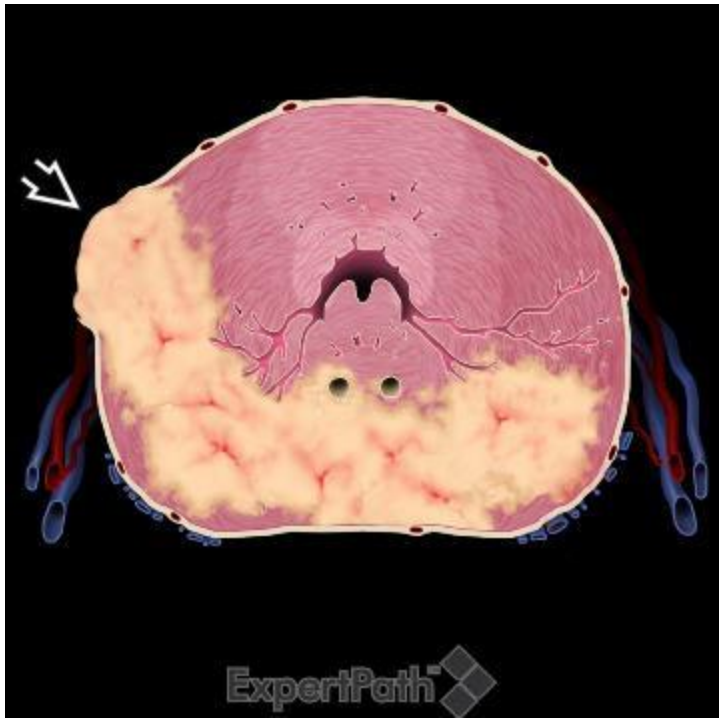
Pattern 5 - no glands (solid sheets, single cells, cords, comedonecrosis)

Gleason score	Grade group
Gleason 6 (3+3)	Grade group 1
Gleason 7 (3+4)	Grade group 2
Gleason 7 (4+3)	Grade group 3
Gleason 8	Grade group 4
Gleason 9 or 10	Grade group 5

Tumor pathologic staging



Stage pT2: PCa is organ confined.



Stage pT3a: extraprostatic disease.



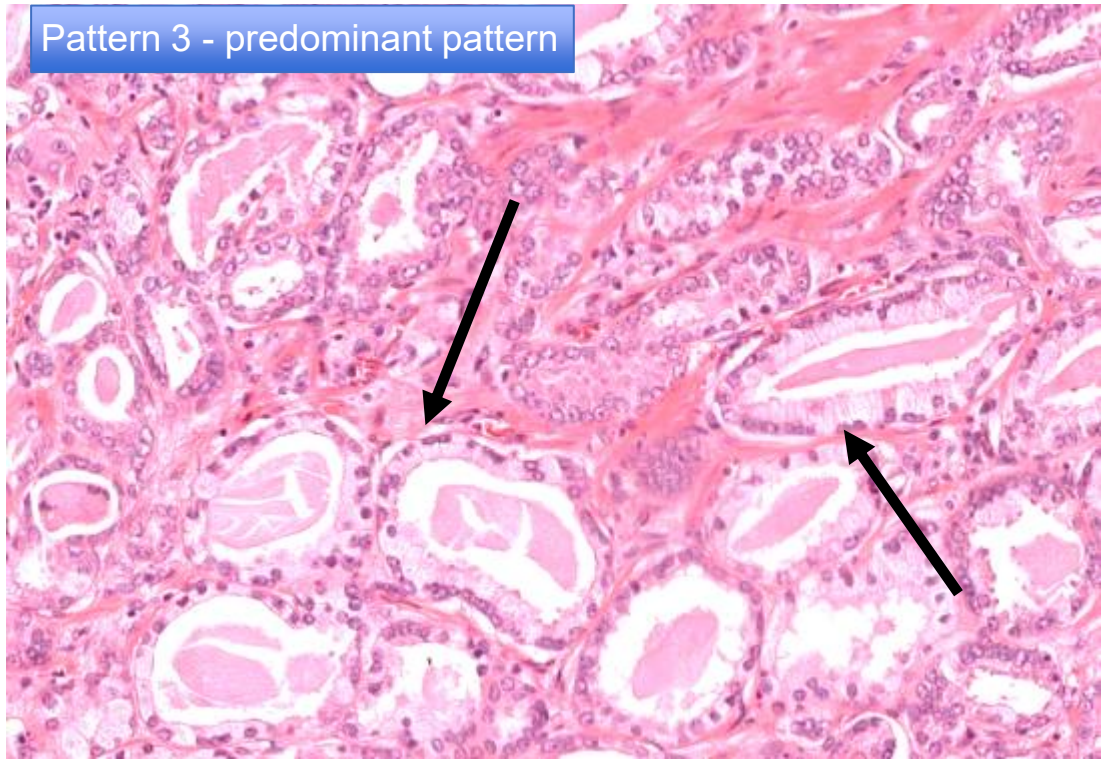
Stage pT3b: seminal vesicle involvement

Pathology report of a radical prostatectomy specimen

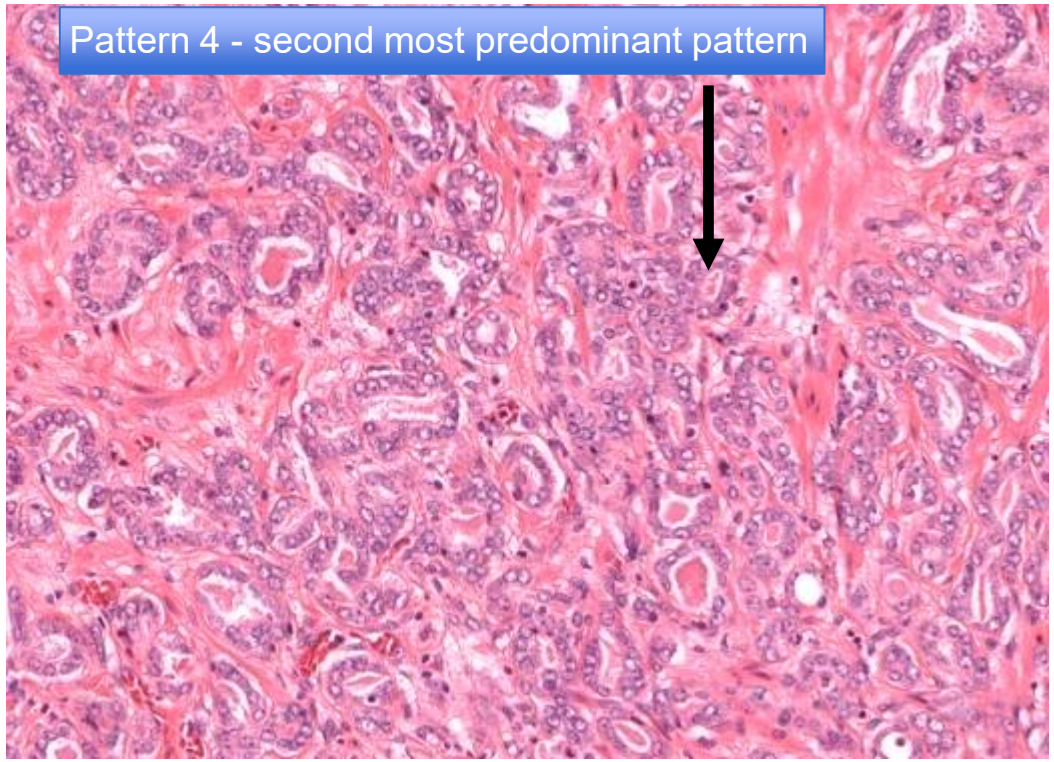
- Grade of tumor: Gleason histologic grading (Gleason score)
- Extent local spread: extraprostatic tissue, bladder neck, seminal vesicle, surgical margins
- Status of lymph nodes
- Percent of tissue involvement by tumor

CASE 3 - Histology of prostatic adenocarcinoma

Well-formed separate infiltrating malignant glands



Fused, complex glands that cannot easily separated

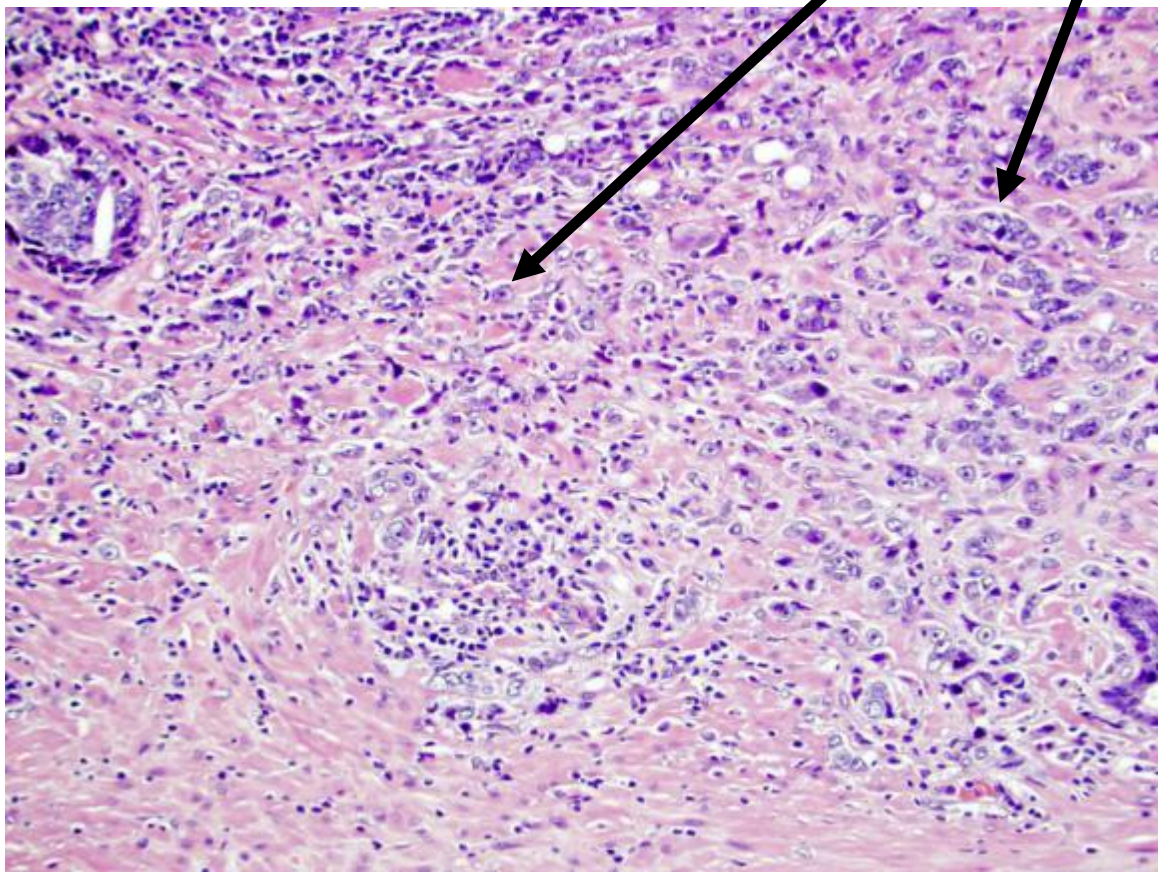


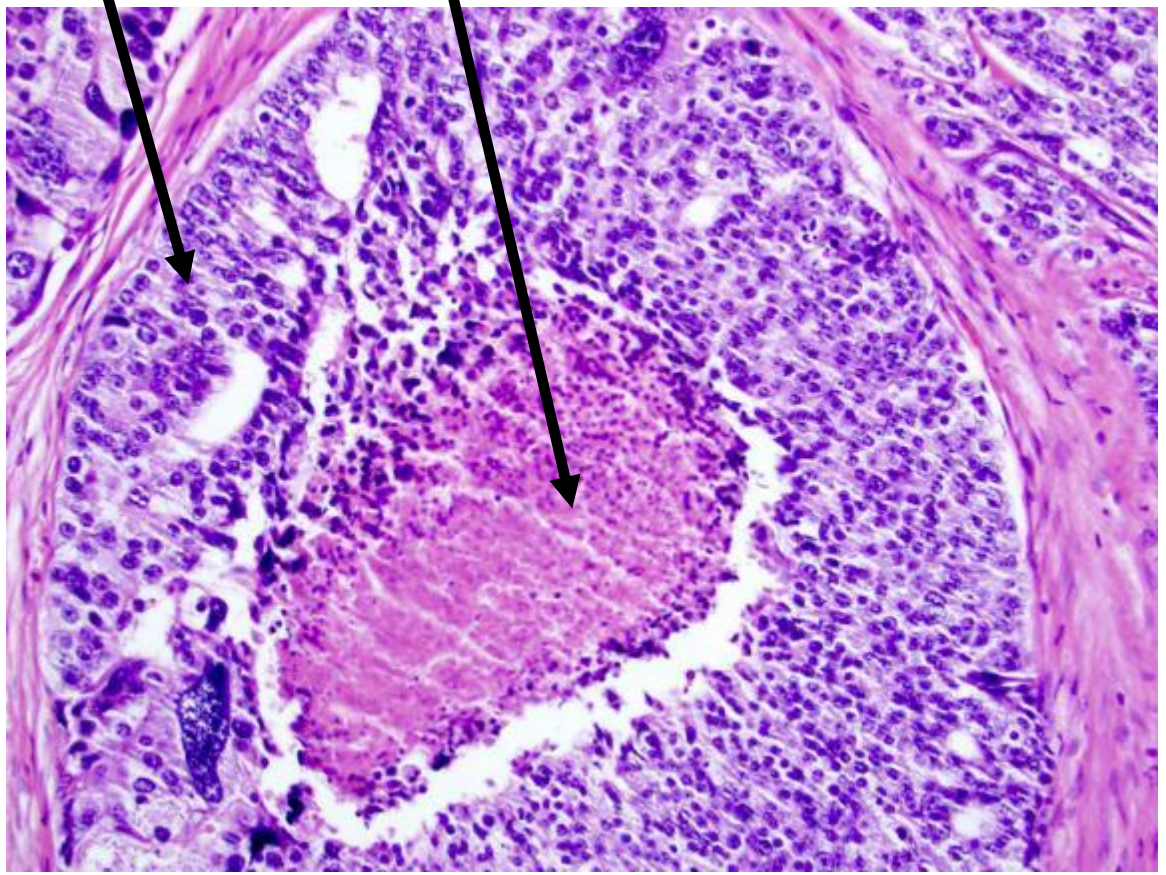
Gleason score 3 + 4 = 7 (intermediate grade)

CASE 3 - Histology of prostatic adenocarcinoma

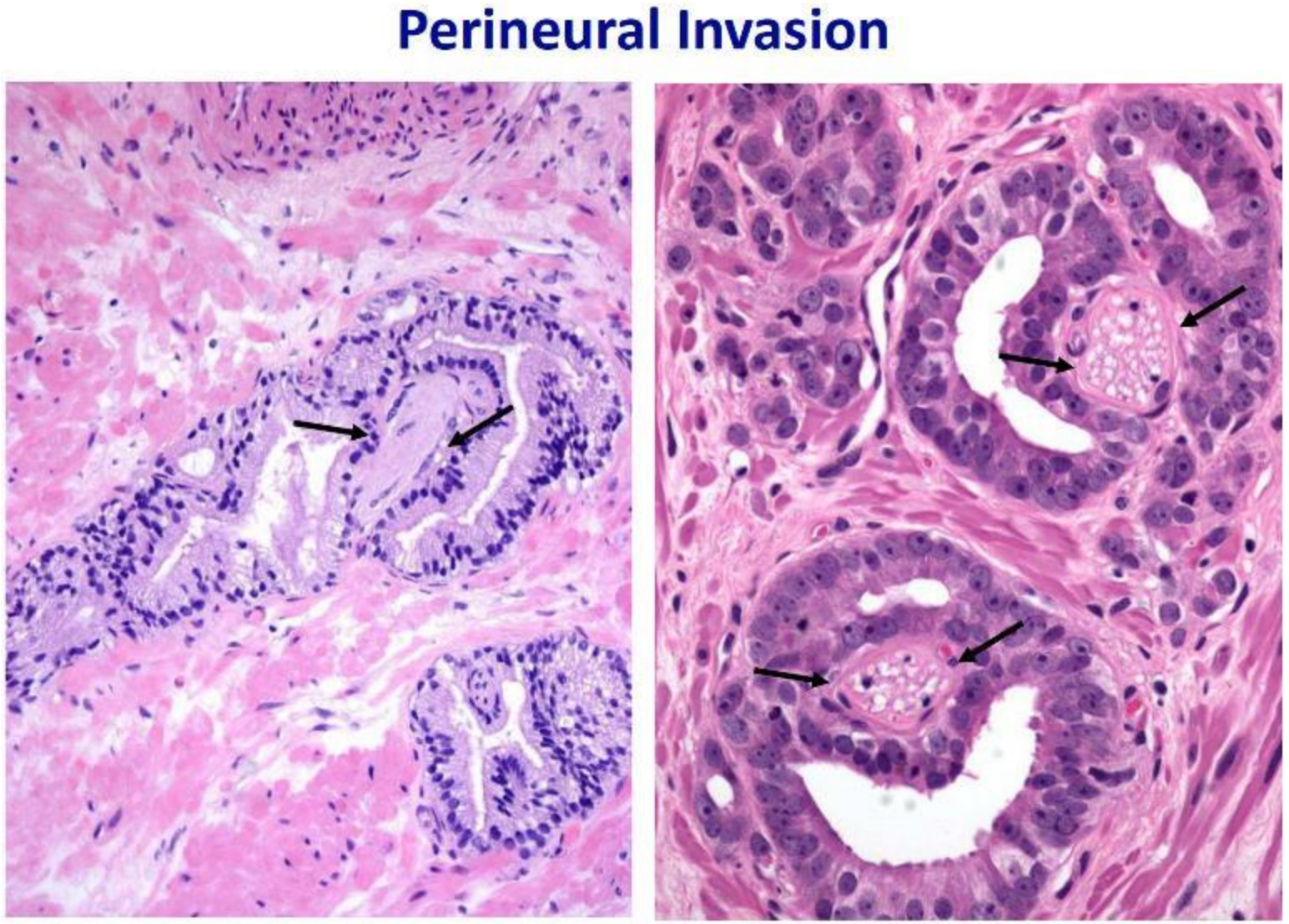
Pattern 5 – not seen in *this* patient's case

No glandular differentiation: single cells, cords, solid growth, comedonecrosis





Tumor wrapping around nerves is perineural invasion



- Perineural invasion is common in prostate carcinoma. It is not an independent prognostic factor.
- However, extensive intraglandular perineural invasion may indicate perineural invasion outside of the prostate capsule

Other prostate malignancies

- Acinar adenocarcinoma is the most common malignant neoplasm of the prostate
- Another type of epithelial neoplasm is recognized; namely, ductal carcinoma
- Also, the prostate stroma may be affected by mesenchymal tumors such as stromal sarcoma and leiomyosarcoma

Case 4

CASE 4 - Clinical history

- A 33-year-old patient was admitted for a left testicular mass found on self-examination; there was vague tenderness
- Testing for serum tumor markers was performed; alpha-fetoprotein (AFP) was normal; lactate dehydrogenase (LDH) was elevated and β-hCG was mildly elevated
- Ultrasound exam showed a 5.0 cm solid mass, well-circumscribed and confined to the testis
- There was no epididymal or spermatic cord involvement, and no inguinal lymphadenopathy
- An orchiectomy was performed
- The pathology laboratory received the specimen, consisting of an intact testis and spermatic cord

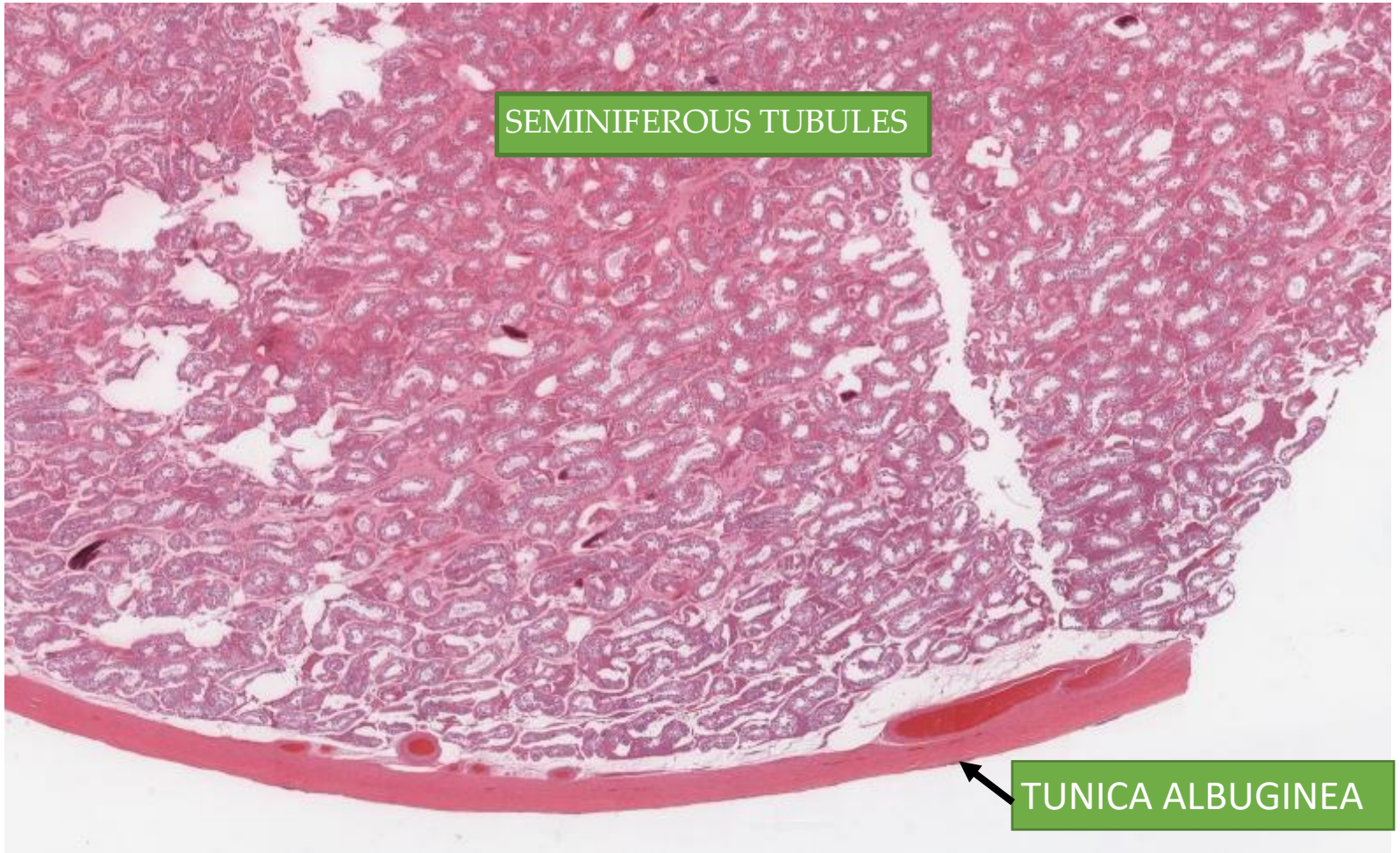
CASE 4 – GROSS SPECIMEN - Below is an image of the surgical specimen, bisected.



CASE 4: Key features of clinical history

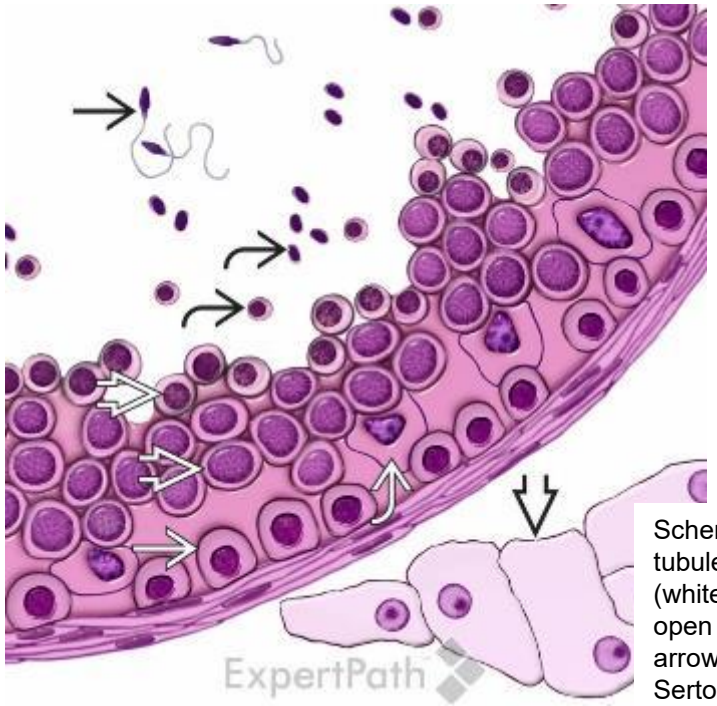
- **Age:**
  - Testicular germ cell tumors (TGCT) are most common in young adults (20–40 yrs)
    - Seminomas affect older individuals compared with nonseminomatous germ cell tumors
    - On average, patients with mixed germ cell tumors are ~10 years younger at diagnosis than those with pure seminoma
    - Yolk sac tumor is the most common TGCT in children (mainly seen in children < 3 years)
- **Associated symptoms:**
  - Seminomas are typically painless or vaguely painful masses
  - Gynecomastia may be seen in patients with choriocarcinoma

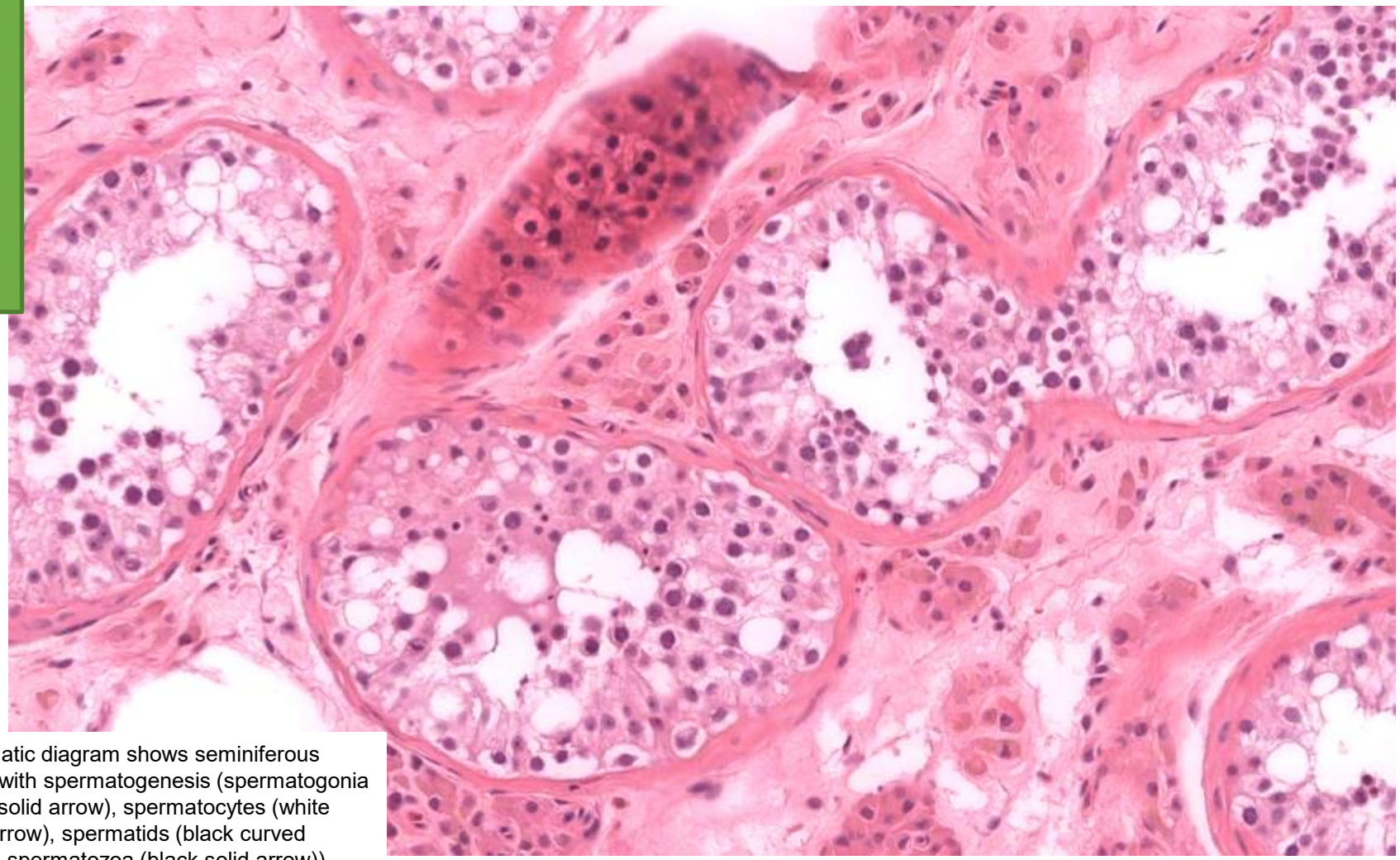
CASE 4 – MICROSCOPY – NORMAL TESTIS



CASE 4 - NORMAL TESTIS

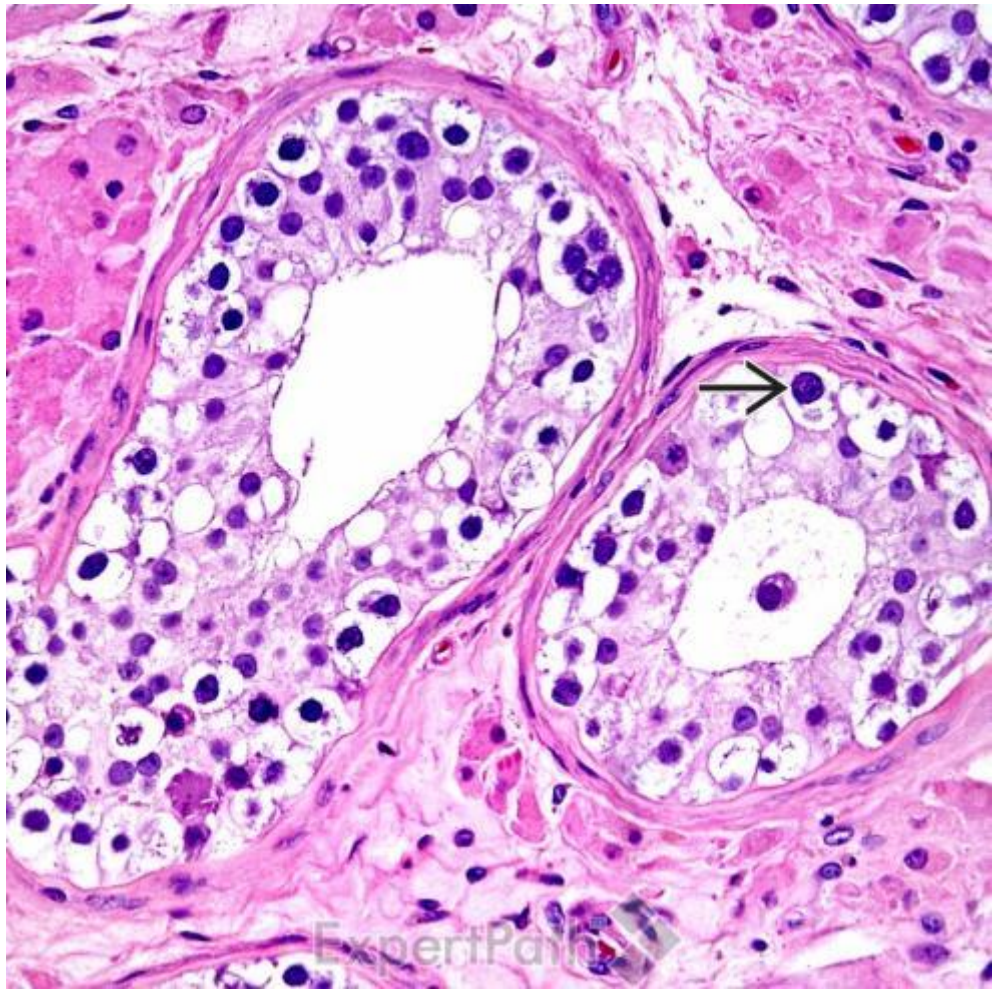
- The cells of the seminiferous tubules are:
- Sertoli cells
- Primary spermatocytes
- Secondary spermatocytes
- Spermatids



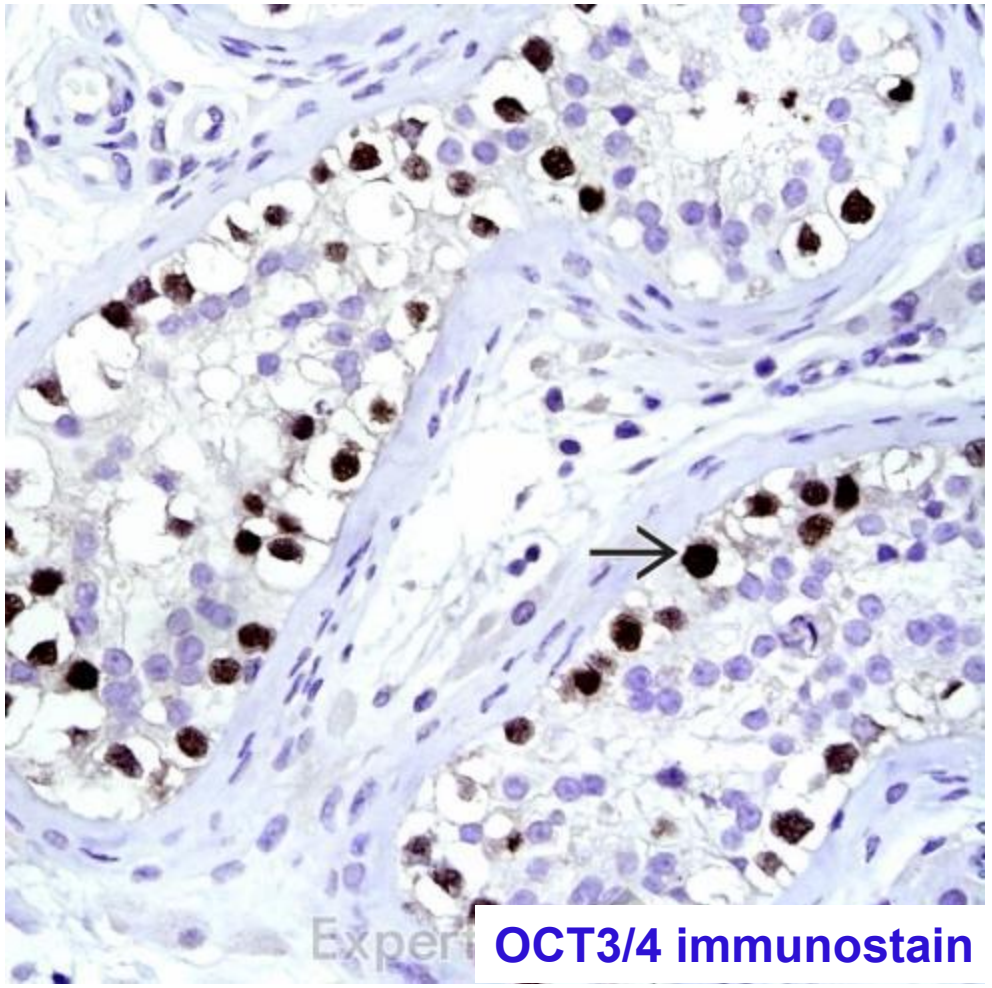


Schematic diagram shows seminiferous tubule with spermatogenesis (spermatogonia (white solid arrow), spermatocytes (white open arrow), spermatids (black curved arrow), spermatozoa (black solid arrow)). Sertoli cells (white curved arrow) and Leydig cells (black open arrow) are also shown.

Germ cell neoplasia in situ (GCNIS)

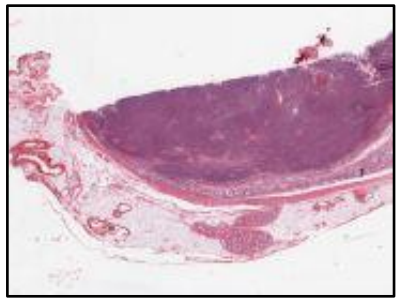


GCNIS is characterized by large atypical cells located along the periphery of the tubules (spermatogonial niche) (black solid arrow). The cells have centrally located nuclei, prominent nucleoli, and abundant clear cytoplasm.

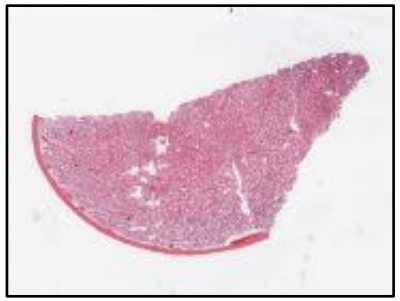


Case 4A

As the genitourinary pathologist on service, you review the patient's orchiectomy specimen, with the slide linked below:



Bladder, Prostate, Testis VS 4-1



normal for comparison
(no annotations)

Case 4A

Examine Bladder, Prostate, Testis VS 4-1 and answer the following questions:

Q1: Examine the normal seminiferous tubules shown in the black box. Thinking back to the *Histology of the Testis & Sperm Release Pathway* histology lab, what are the major cell types and gametogenic/developmental stages seen in these tubules?

Spermatogonia, spermatocytes (primary and secondary), and spermatids.

Q2: What is this (normal) structure in the blue circle seen adjacent to the testis in this resection specimen?

This is the epididymis.

Case 4A

Continue to examine Bladder, Prostate, Testis VS 4-1 and answer the following questions:

Q3: Examine the appearance of the patient's tumor cells shown in the white box. Do the tumor cells look similar to any of the normal cell types in the testis? How are they organized within the tumor?

The tumor cells look similar to immature germ cells. They are organized into sheets or cords of tumor cells with some fine vessels between them.

Q4: What type of inflammatory cell is seen percolating through the tumor; examples of which are shown within the yellow circle?

These are lymphocytes.

Case 4A

Continue to examine Bladder, Prostate, Testis VS 4-1 and answer the following questions:

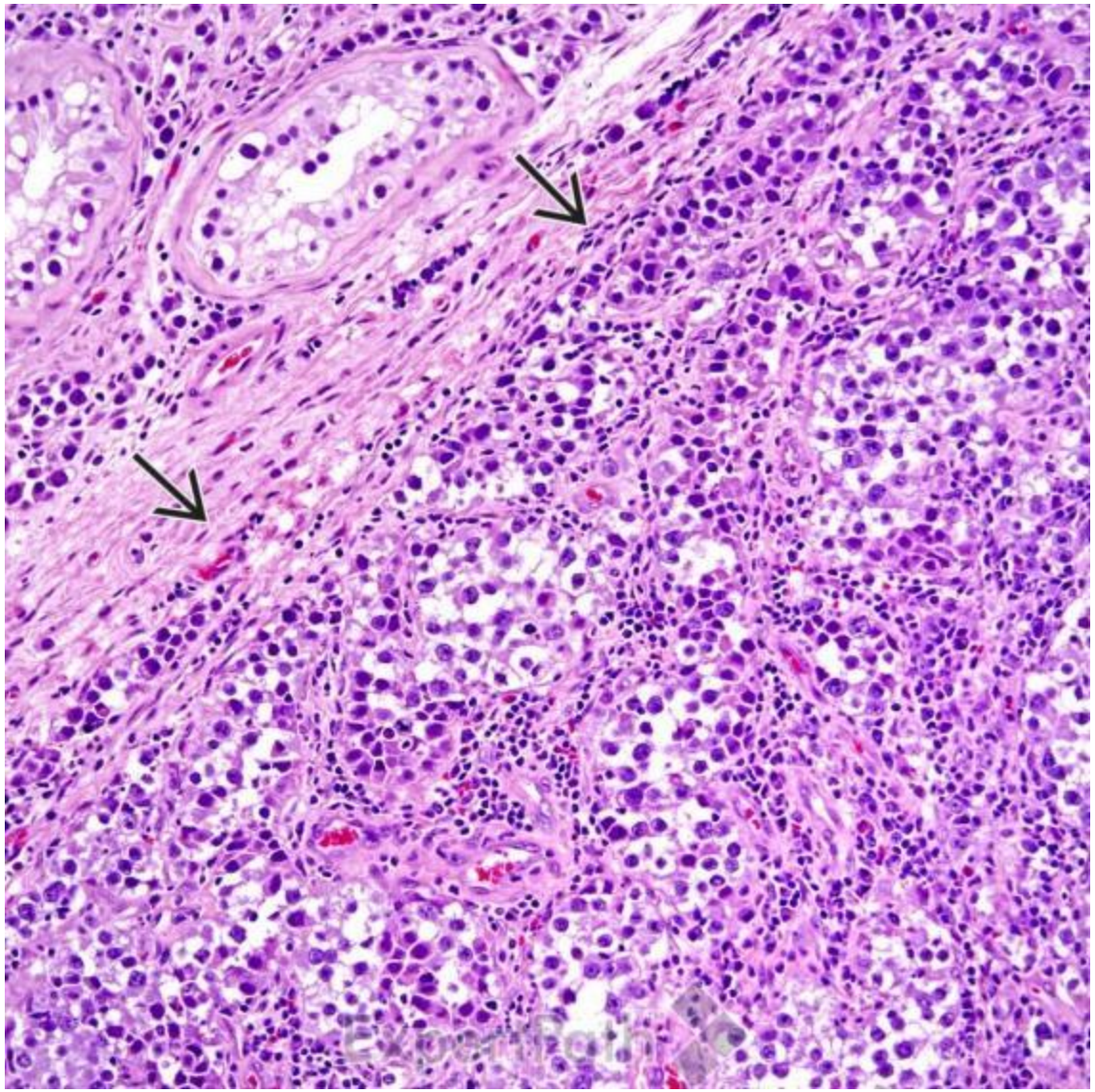
Q5: The eosinophilic region in the green, dashed box shows some cellular outlines without nuclei and some reactive inflammatory cells. What process is happening here? What does this tell us about the biology of the patient's neoplasm?

This is necrosis, which indicates that the tumor growth is outrunning its blood supply.

Q6: The patient's tumor is surrounded by the orange enclosure. Based on the histopathology and epidemiology of testicular cancer generally, what is the most likely tumor type in this patient? Is this patient's tumor restricted to the testis or invades outside of it?

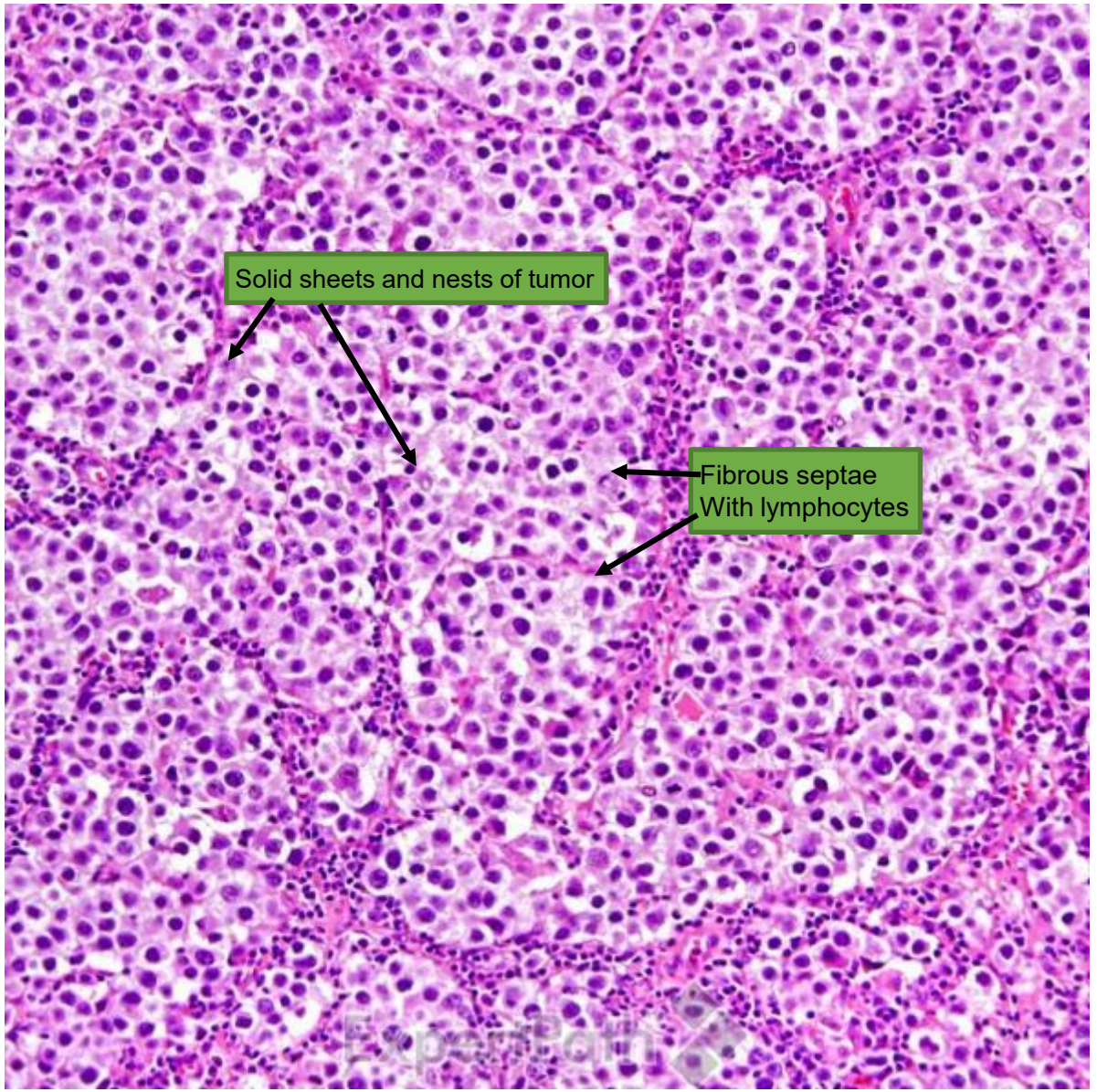
This is most likely a seminoma. The tumor is restricted to the testis.

CASE 4A – PATIENT'S TUMOR



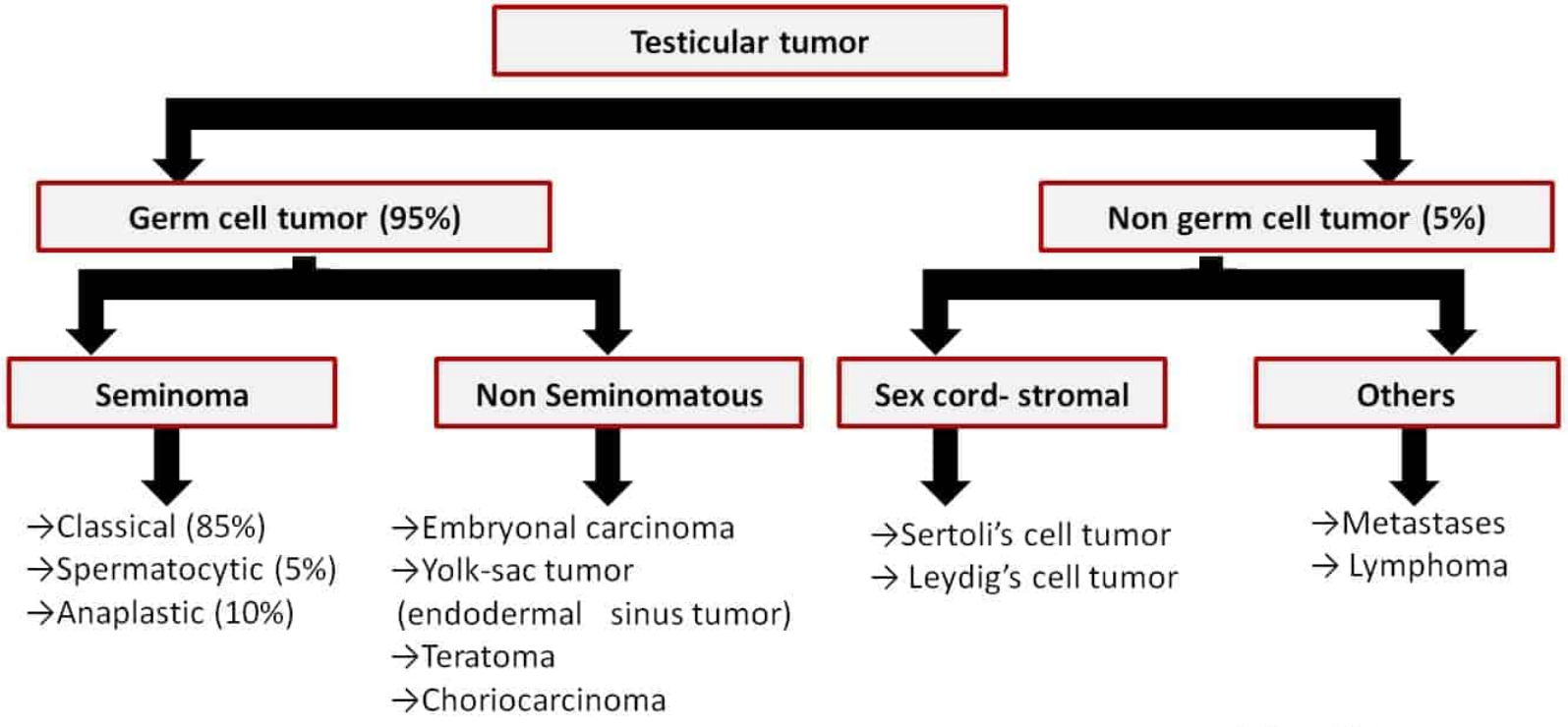
CASE 4A – PATIENT'S TUMOR

- Nests and sheets of cells, with thin fibrous septa separating nests and lobules of tumor cells
- Within the septa are variably prominent lymphocytes
- The tumor cells are rather monotonous with clear to pale to eosinophilic cytoplasm



TESTICULAR TUMOR TYPES

- Germ cell tumors account for 95% of all testicular tumors
- TGCTs are divided into seminomatous (50%) and nonseminomatous categories
- TGCTs can be pure or mixed
- Non-Hodgkin lymphoma is the commonest tumor in men over 60



What is the most likely diagnosis?

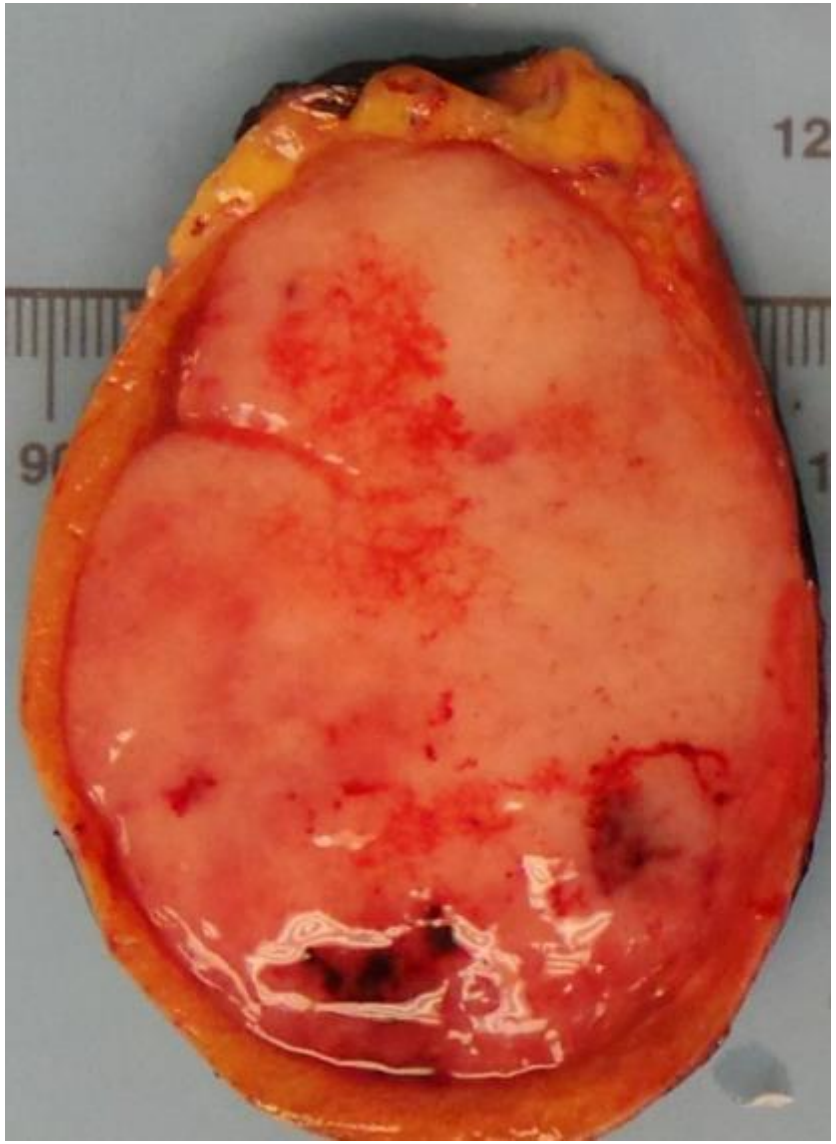
A. Testicular torsion
B. Metastatic adenocarcinoma of the prostate gland
C. Large B-cell lymphoma
D. Tuberculosis
E. Seminoma

What is the most likely diagnosis?

A. Testicular torsion
B. Metastatic adenocarcinoma of the prostate gland
C. Large B-cell lymphoma
D. Tuberculosis
E. Seminoma

Lymphomas may involve the testis, usually when there is systemic involvement by a high-grade lesion. Lymphomas do not elaborate hormones.

CASE 4 – GROSS SPECIMEN



- Seminoma is a solid, fleshy tan to yellow mass which is frequently well-circumscribed and may have small foci of hemorrhage or necrosis
- The tumor is confined to the testicular parenchyma and there are no satellite lesions
- The tumor does not invade the tunica or epididymis

SEMINOMA – Clinical features

- Seminoma is the most common germ cell tumor, approximately 50% of all TGCTs
- Seminoma occurs in older patients when compared with NSGCT
- Patients typically present with a painless or vaguely painful mass
- Bilateral disease is rare (<2%)
- The outcome for pure seminoma is extremely good, with greater than 95% survival at 5 years due to its radiosensitive and chemosensitive nature

Which one of the following is NOT a major type of epithelial neoplasm of the testis?

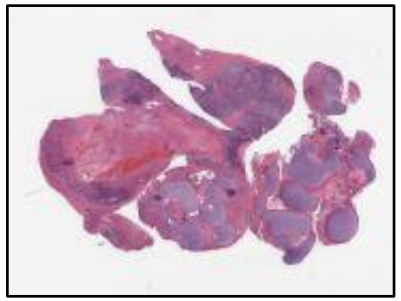
A. Yolk sac tumor
B. Choriocarcinoma
C. Seminoma
D. Leiomyosarcoma
E. Teratoma
F. Embryonal carcinoma

Which one of the following is NOT a major type of epithelial neoplasm of the testis?

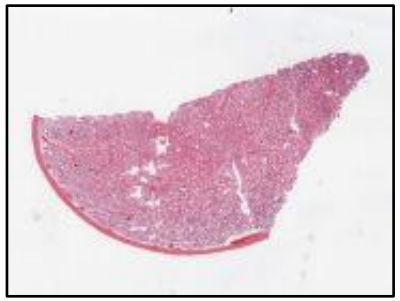
A. Yolk sac tumor
B. Choriocarcinoma
C. Seminoma
D. Leiomyosarcoma
E. Teratoma
F. Embryonal carcinoma

Case 4B

A different patient's orchiectomy specimen is linked below. Compare the morphology of this patient's tumor to that of the original patient:



Bladder, Prostate, Testis VS 4-2



normal for comparison
(no annotations)

Case 4B

Examine Bladder, Prostate, Testis VS 4-2 and answer the following questions:

Q1: The region shown in the red box is best classified as embryonal carcinoma. How would you describe the growth pattern and cellular appearance of this tumor type?

It is a very cellular tumor with a high nucleus-to-cytoplasmic ratio and indistinct cell borders, growing in sheets.

Q2: What tissue type is shown within the black circle? (You have not formally covered it yet, but will in the Musculoskeletal Block.) The presence of this tissue in the testis suggests what additional tumor type?

This is cartilage, suggesting a teratoma (which is composed of tissues from more than one embryonic germ layer).

Case 4B

Continue to examine Bladder, Prostate, Testis VS 4-2 and answer the following questions:

Q3: Considering this patient's specimen as a whole, it demonstrates two very different morphologies. How, then, is their tumor best classified? What other morphologies may be seen in association with this tumor type?

This is a mixed (non-seminomatous) germ cell tumor.

Choriocarcinoma and yolk sac tumor may also be seen in a mixed germ cell tumor.

Serum markers in testicular germ cell tumors

- **Serum markers aid diagnosis, prognosis and disease management**
- It is incorporated into the TNM staging
- Serum tumor markers should be obtained:
  - before orchiectomy
  - following orchiectomy
  - during treatment to assess response
  - during surveillance to monitor for recurrence
- **3 serum markers are of importance**
  - HCG (human chorionic gonadotropin)
  - AFP (alpha fetoprotein)
  - LDH (lactate dehydrogenase)

SERUM TUMOR MARKERS IN TESTICULAR GERM CELL TUMORS

1. **Human chorionic gonadotropin (hCG):**
   - Modest elevation can occur when a seminoma or embryonal carcinoma contains some syncytial giant cells
   - Choriocarcinomas secrete high levels of hCG but no AFP

2. **α-fetoprotein (AFP):**
   - Yolk sac tumor is almost always associated with elevated serum AFP
   - Other nonseminomatous tumors (embryonal carcinoma, teratoma) may also have high AFP
   - Significant elevation of the α-fetoprotein (AFP) level never occurs with pure seminomas

3. **Lactate dehydrogenase (LDH):**
   - Elevated in 50% of men with TGCTs – related to tumor burden:
     - If >2000 U/L – bulky disease
     - May be the only abnormal marker in persistent or recurrent disease - 10%

TESTICULAR GERM CELL TUMOR STAGING

- TNM (tumor, node, and metastasis) of malignant tumors is a standardized and widely accepted staging system
- Size (3 cm)
- Extent of local spread (epididymis, tunica albuginea, hilum, spermatic cord, local lymphatics, scrotum)
- Lymph node status (number; size of largest involved lymph node; extranodal extension)
- Presence of distant metastasis
- Serum markers (S category)

COLLEGE of AMERICAN PATHOLOGISTS

Protocol for the Examination of Radical Orchiectomy Specimens From Patients With Malignant Germ Cell and Sex Cord-Stromal Tumors of the Testis

pT Category

___ pT not assigned (cannot be determined based on available pathological information)

___ pT0: No evidence of primary tumor

___ pTis: Germ cell neoplasia *in situ*

pT1: Tumor limited to testis (including rete testis invasion) without lymphovascular invasion

Subclassification of pT1 applies only to pure seminoma.

___ pT1a: Tumor smaller than 3 cm in size#

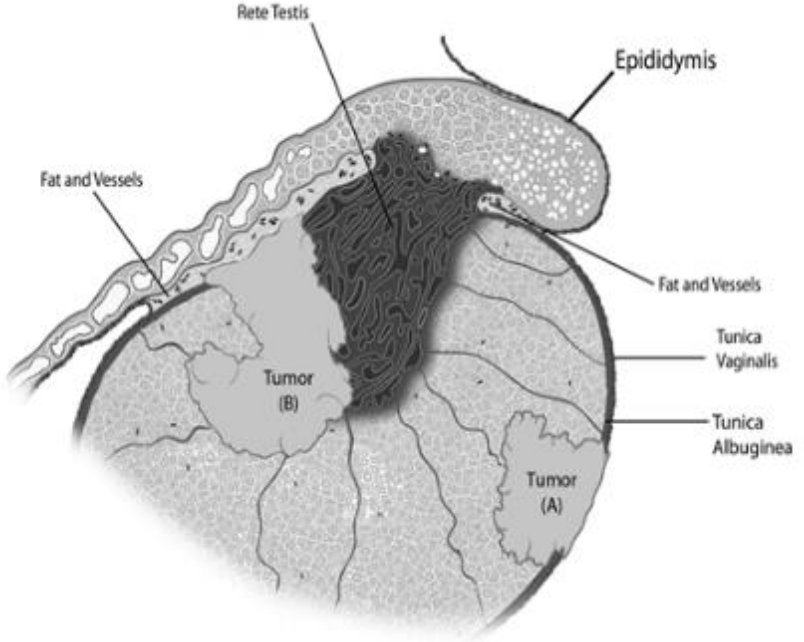
___ pT1b: Tumor 3 cm or larger in size#

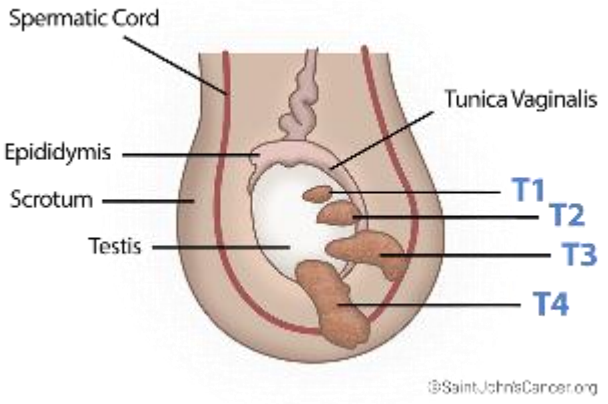
___ pT1 (subcategory cannot be determined)

___ pT2: Tumor limited to testis (including rete testis invasion) with lymphovascular invasion, or tumor invading hilar soft tissue or epididymis or penetrating visceral mesothelial layer covering the external surface of tunica albuginea with or without lymphovascular invasion

___ pT3: Tumor directly **invades spermatic cord soft tissue** with or without lymphovascular invasion

___ pT4: Tumor invades scrotum with or without lymphovascular invasion





Which of the following is true regarding staging of testicular tumors?

A. Staging is optional
B. Staging is determined by age, tumor size and type of tumor
C. Staging is not done
D. Staging is determined by tumor size, node status and metastasis
E. Staging is determined by gender, age, and type of tumor

Which of the following is true regarding staging of testicular tumors?

A. Staging is optional
B. Staging is determined by age, tumor size and type of tumor
C. Staging is not done
D. Staging is determined by tumor size, node status and metastasis
E. Staging is determined by gender, age, and type of tumor

Prognostic Factors

- 90-95% overall survival rate in USA for germ cell tumors with surgery, chemotherapy and radiation therapy
- Depends on histologic type, stage, and treatment
- Most types have favorable prognosis and respond well to tumor-specific chemotherapy and/or radiation therapy
- Morphologic prognostic factors
    - Proportion of embryonal carcinoma (> 80%, poor prognosis)
    - Choriocarcinoma (pure or as mixed) imparts a poor prognosis
    - Proportion of teratoma (> 50% favorable prognosis)
    - Others: Tumor size (> 4 cm), lymphovascular invasion
- Serum tumor marker levels
    - Patients who have pure seminoma in orchiectomy specimen with elevated AFP should be considered as NSGCT and extensive sampling is recommended
    - Levels of LDH elevation correlate with tumor burden
    - Markedly increased hCG associated with poor prognosis